



SYLLABUS

(Choice Based Credit System)

Faculty of Technology & Engineering

Bachelor of Technology Programme
(Second Year Mechanical Engineering)



Education Campus – Changa, (ECC), hitherto a conglomerate of institutes of professional education in Engineering, Pharmacy, Computer Applications, Management, Applied Sciences, Physiotherapy and Nursing, is one of the choicest destinations by students. It has been transformed into Charotar University of Science and Technology (CHARUSAT) through an Act by Government of Gujarat. CHARUSAT is permitted to grant degrees under Section-22 of UGC- Govt. of India.

The journey of CHARUSAT started in the year 2000, with only 240 Students, 4 Programmes, one Institute and an investment of about Rs. 3 Crores (INR 30 million). At present there are seven different institutes falling under ambit of six different faculties. The programmes offered by these faculties range from undergraduate (UG) to Ph.D degrees including M.Phil. These faculties, in all offer 23 different programmes. A quick glimpse in as under:

Faculty	Institute	Programmes Offered
Faculty of Technology & Engineering	Chandubhai S. Patel Institute of Technology	B. Tech M. Tech Ph. D
	Devang Patel Institute of Advance Technology and Research	B.Tech CE CSE IT
Faculty of Pharmacy	Ramanbhai Patel College of Pharmacy	B. Pharm M. Pharm Ph. D PGDCT/ PGDPT
Faculty of Management Studies	Indukaka Ipcowala Institute of Management	M.B.A PGDM Ph.D Dual Degree BBA+MBA
Faculty of Applied Sciences	P.D.Patel Institute of Applied Sciences	M.Sc M.Phil Ph.D Dual Degree B.Sc+M.Sc

Faculty	Institute	Programmes Offered
Faculty of Computer Applications	Smt. Chandaben Mohanbhai Patel Institute of Computer Applications	M.C.A/MCA (Lateral) M.Sc IT Ph. D Dual Degree BCA+MCA
Faculty of Medical Sciences	Ashok and Rita Institute of Physiotherapy	B.PT M.PT Ph.D
	Manikaka Topawala Institute of Nursing	B.Sc M.Sc GNM
	Charotar Institute of Paramedical Sciences	Ph.D PGDHA

The development and growth of the institutes have already led to an investment of over Rs.63 Crores (INR 630 Million). The future outlay is planned with an estimate of Rs. 250 Crores (INR 2500 Million).

The University is characterized by state-of-the-art infrastructural facilities, innovative teaching methods and highly learned faculty members. The University Campus sprawls over 100 acres of land and is Wi-Fi enabled. It is also recognized as the Greenest Campus of Gujarat.

CHARUSAT is privileged to have 350 core faculty members, educated and trained in Stanford, IITs, IIMs and leading Indian Universities, and with long exposure to industry. It is also proud of its past students who are employed in prestigious national and multinational corporations.

From one college to the level of a forward-looking University, CHARUSAT has the vision of entering the club of premier Universities initially in the country and then globally. High Moral Values like Honesty, Integrity and Transparency which has been the foundation of ECC continues to anchor the functioning of CHARUSAT. Banking on the world class infrastructure and highly qualified and competent faculty, the University is expected to be catapulted into top 20 Universities in the coming five years. In order to align with the global requirements, the University has collaborated with internationally reputed organizations like Pennsylvania State University – USA, University at Alabama at Birmingham – USA, Northwick Park Institute –UK, ISRO, BARC, etc.

- ☞ Participatory and interactive discussion-based classes.
- ☞ Sessions by visiting faculty members drawn from leading academic institutions and industry.
- ☞ Regular weekly seminars.
- ☞ Distinguished lecture series.
- ☞ Practical, field-based projects and assignments.
- ☞ Summer training in leading organizations under faculty supervision in relevant programmes.
- ☞ Industrial tours and visits.
- ☞ Extensive use of technology for learning.
- ☞ Final Placement through campus interviews.

Exploration in the field of knowledge through research and development and comprehensive industrial linkages will be a hallmark of the University, which will mould the students for global assignments through technology-based knowledge and critical skills.

The evaluation of the student is based on grading system. A student has to pursue his/her programme with diligence for scoring a good Cumulative Grade Point Average (CGPA) and for succeeding in the chosen profession and life.



CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (Mechanical Engineering) Programme
(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
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Year 2024-25

CHARUSAT

FACULTY OF TECHNOLOGY AND ENGINEERING

ACADEMIC REGULATIONS

Bachelor of Technology Programmes

Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

(i)	Undergraduate programme (B.Tech)
Minimum	8 semesters (4 academic years)
Maximum	16 semesters (8 academic years)

3. Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – I attached

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment. The continuous assessment will be conducted by the respective department/institute.

7.1.2 Final end-semester examination by the University through written paper or practical test or oral test or presentation by the student or combination of these.

7.1.3 The weightages of continuous assessment and end-semester University examination in overall assessment shall depend on individual course as approved by Academic Council through Board of Studies.

7.1.4 The performance of candidate in continuous assessment and end-semester examination together (if applicable) shall be considered for deciding the final grade in a course.

7.1.5 In order to earn the credit in a course a student has to obtain grade other than FF.

7.2 Performance in continuous assessment and end-semester University Examinations

7.2.1 Minimum performance with respect to continuous assessment as well as end-semester university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations are as follows:

Minimum percentage marks to be obtained in end-semester University examination (for applicable courses)	Minimum overall percentage marks to be obtained in each course
40%	45%

7.2.2 If a candidate obtains minimum required percentage of marks in end-semester University examination in an applicable course but fails to obtain minimum required overall percentage of marks, he/she has to repeat the examination till the minimum required overall percentage of marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	≥80	≥73 <80	≥66 <73	≥60 <66	≥55 <60	≥50 <55	≥45 <50	<45
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) $SGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses in the semester

(ii) $CGPA = \frac{\sum C_i G_i}{\sum C_i}$ where C_i is the number of credits of course i
 G_i is the Grade Point for the course i
 and $i = 1$ to n , n = number of courses of all semesters up to which CGPA is computed.

9. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	CGPA Range
First Class with Distinction	$CGPA \geq 7.5 \ \& \ < 10.0$
First class	$CGPA \geq 6.0 \ \& \ < 7.5$
Second Class	$CGPA \geq 5.0 \ \& \ < 6.0$
Pass Class	$CGPA \geq 5.0$

Grade sheets of only the final semester shall indicate the class. In case of all the other semesters, it will simply indicate as Pass / Fail.

10. Detention Criteria

- No student will be allowed to move further in next semester if CGPA is less than 3 at the end of an academic year.
- A Student will not be allowed to move to third year if he/she has not cleared all the courses of first year.
- A student will not be allowed to move to fourth year if he/she has not cleared all the courses of first and second year.

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
(CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM
FOR
BACHELOR OF TECHNOLOGY & ENGINEERING

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 4 types of courses: Foundation courses, Core courses, Elective courses and Non-credit (audit) courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or

which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Institute Elective Course (IE)

Institute elective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

D. Cluster Elective Course (CE):

An 'Elective Course' is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.4. Non Credit Course (NC) - AUDIT Course

A 'Non Credit Course' is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student's Grade Sheet but the grade of the course will not be consider to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5. Medium of Instruction

The Medium of Instruction will be English.

Department of Mechanical Engineering – Chandubhai S Patel Institute of Technology

Vision

To be recognized as a center for outstanding education and research in field of mechanical engineering.

Mission

To produce well qualified engineers, who are innovative, contributors to their profession by catering to diverse societal needs.

Program Educational Objectives:

The objectives of the Mechanical Engineering Undergraduate Program are to produce graduate who can:

1. Apply their knowledge and expertise to find solutions of the problems in core and allied areas of engineering, exhibiting creativity and entrepreneurial skills.
2. Conduct themselves in an ethical and professional manner, exhibiting effective skills in communication, management, teamwork and leadership.
3. Continue their intellectual development through self-study and continuing education, practicing life-long learning.

Program Outcomes: Mechanical Engineering

- PO1 **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- PO2 **Problem analysis:** Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PO3 **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- PO4 **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

- PO5 **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
- PO6 **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- PO7 **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- PO8 **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- PO9 **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- PO10 **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- PO11 **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- PO12 **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes: Mechanical Engineering

- PSO1 The Mechanical Engineering graduates will be able to analyse, design, and evaluate the performance of mechanical components and systems by using various tools.
- PSO2 The Mechanical Engineering graduates will be able to plan, automate and evaluate manufacturing and system processes.

TEACHING & EXAMINATION SCHEME
B. TECH. PROGRAMME IN MECHANICAL ENGINEERING

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY (CHARUSAT)							(w.e.f. June 2024)				
TEACHING & EXAMINATION SCHEME FOR SECOND YEAR B. TECH. PROGRAMME IN MECHANICAL ENGINEERING											
Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Interna I	External	Interna I	External	
SY Sem-3	MA251	Engineering Mathematics III	4	0	4	4	30	70	00	00	100
	CL243	Mechanics of Solids	4	2	6	5	30	70	25	25	150
	ME242	Engineering Thermodynamics	3	0	3	3	30	70	00	00	100
	ME253	Mechanisms & Machines	4	2	6	5	30	70	25	25	150
	ME251	Casting & Joining Processes	3	2	5	4	30	70	25	25	150
	XXXXX	University Elective - I	2		2	2	00	00	30	70	100
	HS121.02 A	Creativity, Problem Solving And Innovation	2		2	2	00	00	30	70	100
			28			25					800
SY Sem-4	ME252	Machining & Forming Processes	3	2	5	4	30	70	25	25	150
	ME246	Material Engineering & Metallurgy	4	2	6	5	30	70	25	25	150
	ME247	Fluid Mechanics	3	2	5	4	30	70	25	25	150
	ME250	Dynamics of Machines	4	2	6	5	30	70	25	25	150
	MA249	Numerical Methods	3	2	5	4	30	70	25	25	150
	HS111.03 A/ HS113 A	Human Values and Ethics/ Indian Knowledge System	2		2	2	00	00	30	70	100
	XXXXX	University Elective - II	2		2	2	00	00	30	70	100
			31			26					900

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

MA – 251 ENGINEERING MATHEMATICS – III

3rd Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	0	4	4
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Laplace Transform and Its Applications	15
2.	Fourier Series and Its Applications	15
3.	Vector Differential Calculus	10
4.	Vector Integral Calculus	12
5.	Curve Fitting	08

Total hours (Theory): 60

Total hours (Lab): 00

Total: 60

B. Detailed Syllabus:

1	Laplace Transform and Its Applications	15 Hours	25%
1.1	Laplace transform as an improper integral and its existence.		
1.2	Laplace transform of elementary functions and its properties.		
1.3	Inverse Laplace transform and its properties.		
1.4	First and second shifting theorems. Laplace transform of derivatives and integrals.		
1.5	Convolution theorem and its application to obtain inverse Laplace transform.		
1.6	Laplace transform of periodic functions, Unit step function, Unit impulse function (Dirac delta function).		
1.7	Solving differential equations using Laplace transform.		
1.8	Applications of Laplace transform to ODE: Mechanical vibration system, deflection of beams.		
2	Fourier Series and Its Applications	15 Hours	25%
2.1	Periodic functions, Dirichlet's conditions, Trigonometric series.		

2.2	Euler formulae, Fourier series of periodic function of period p .		
2.3	Fourier series of Discontinuous functions..		
2.4	Fourier series of Even and odd functions, half range Fourier series.		
2.5	Solutions of heat, wave, Laplace equations by method of separation of variables and Fourier series.		
3	Vector Differential Calculus	10 Hours	17%
3.1	Revision of concepts of Vector algebra, Scalar and Vector fields.		
3.2	Gradient of a scalar functions, Directional derivatives.		
3.3	Divergence and Curl of a vector field and their properties.		
3.4	Physical interpretations of gradient, divergence and curl. Irrotational and Solenoidal vector fields.		
3.5	Scalar potential function.		
4	Vector Integral Calculus	12 Hours	20%
4.1	Introduction to line integrals and examples.		
4.2	Statement and examples of Green's theorem.		
4.3	Introduction to Surface and Volume integrals.		
4.4	Statements and examples of Stokes' and Gauss' divergence theorem.		
5	Curve Fitting	08 Hours	13%
5.1	Normal equations.		
5.2	Fitting a straight line: $y = a + bx$ (linear regression).		
5.3	Fitting a parabola: $y = a + bx + cx^2$.		
5.4	Fitting a curve by transformation: $y = ab^x$, $xy^a = b$, $y = ax^b$.		

C. Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject must be discussed.
- Lectures may be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures which carries a 5% component of the overall evaluation.
- Minimum two internal tests/unit tests must be conducted and average of two will be considered as a part of 15% overall evaluation.
- Quizzes (surprise tests) /Oral tests/ Viva/Assignment will be conducted which carries 10% component of the overall evaluation.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Represent continuous time periodic signals using concepts of Fourier series and determine the solution of complex engineering problems heat, wave and Laplace's equations using it.

CO2: Classify and apply the necessary Laplace transform and inverse Laplace transform techniques to solve ordinary differential equations of real world problems.

CO3: Differentiate the dot product, cross product, length of vectors, partial derivatives, derivatives of vector-valued functions, gradient functions and these tools are employed successfully in different branches of engineering and physics.

CO4: Analyse parameterize curves and calculate line integrals using them and Use vector operators and implement into surface integrals, Classify and examine the Green, Stokes and Divergence theorem

CO5: Appraise the Least squares fitting method of data approximation and discuss the concept of curve fitting of different types of curves using specific data.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO2	3	2	1	1	1	-	1	-	-	1	-	1	2	2
CO3	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO4	3	2	2	1	1	-	1	-	-	1	-	1	2	2
CO5	3	3	2	2	2	-	1	-	-	1	-	1	2	2

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Erwin Kreyszig: Advanced Engineering Mathematics, 10th Ed., Jhon Wiley & Sons, India, 1999.
2. Shankar S. Sastry; Introductory methods of numerical analysis. PHI Learning Pvt. Ltd., 2015.

Reference Books:

1. Zafar Ahsaan; Differential equations and their applications. PHI Learning Pvt. Ltd., 2004.
2. H. K. Dass; Advanced engineering mathematics. S. Chand, 2008.
3. Debnath, Lokenath, and Dambaru Bhatta. Integral transforms and their applications. CRC press, 2014.
4. M. D. Greenberg; Advanced Engineering Mathematics, 2nd ed., Pearson. 1998.
5. S. Grewal; Higher engineering mathematics. Khanna Publisher, New Delhi, 1996.
6. Stewart James; Calculus Early Transcendental, 5th Ed., Thomson India, 2007
7. Kenneth A. Stroud and Dexter J. Booth. Advanced engineering mathematics. Palgrave Macmillan, 2011.

7. Wylie and Barrett; Advanced Engineering Mathematics, Mc graw Hill pub., 1982

URL link: Web site: <http://mathworld.wolfram.com> , <http://en.wikipedia.org/wiki/Math>

CL243: MECHANICS OF SOLIDS

3rd Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction	3
2	Simple Stresses and Strains	15
3	Principal Stresses and Strains	10
4	Shear Force and Bending Moment	10
5	Moment of Inertia	6
6	Bending and Shear Stresses in Homogeneous and Composite Beam Sections	10
7	Strain Energy	6

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

1	Introduction	03 Hours	05%
1.1	Introduction to mechanics of deformable bodies		
1.2	Principle of superposition		
1.3	Classification of loaded bar		
1.4	Gradual, sudden, impact and shock loading		
1.5	Mechanical properties of materials		
2	Simple Stresses and Strains	15 Hours	22%
2.1	Stress and types of stress, Strain and types of strain		
2.2	Stress strain Characteristics for ductile and brittle materials		
2.3	Shear stresses and strains, Elasticity, Hook's law		

- 2.4 Axial and shear deformations, Axial force diagram
- 2.5 Bars of varying section, Bars of uniformly varying cross section
- 2.6 Analysis of stress for statically determinate structures and indeterminate structures
- 2.7 Poisson's ratio, Volumetric strain, Biaxial and tri-axial deformations
- 2.8 Elastic constant and relation between three elastic constants
- 2.9 Stresses due to thermal effect

- 3 Principal Stresses and Strains** **10 Hours 18%**
 - 3.1 Introduction, Sign convention
 - 3.2 Transformation of stresses for a state of stresses under axial loading
 - 3.3 Transformation of stresses for plate under biaxial loading
 - 3.4 Principal plane and principal stresses
 - 3.5 Maximum shear stress, Element subjected to principal stresses
 - 3.6 Mohr's circle for stresses on an oblique section of a body subjected to direct stress in one plane and two plane with or without shear stress
 - 3.7 Pure shear

- 4 Shear Force and Bending Moment** **10 Hours 22%**
 - 4.1 Concept of shear force and bending moment
 - 4.2 Sign conventions
 - 4.3 Relation between bending moment, shear force and rate of loading
 - 4.4 Bending moment and shear force diagrams for statically determinate beams subjected to all different types of loading
 - 4.5 Important points for shear force and bending moment diagram.

- 5 Moment of Inertia** **6 Hours 12%**
 - 5.1 Introduction, Radius of gyration
 - 5.2 Parallel axis theorem and perpendicular axis theorem
 - 5.3 Polar moment of inertia
 - 5.4 Moment of inertia by integration
 - 5.5 Moment of inertia of composite areas

- 6 Bending and Shear Stresses in Homogeneous and composite beam sections** **10 Hours 14%**
 - 6.1 Pure bending, Theory of pure bending
 - 6.2 Assumptions and derivation of theory of simple bending

6.3 Neutral axis, moment resistance, section modulus

6.4 Strength of section

6.5 Bending stress in symmetrical, unsymmetrical and composite sections

6.6 Shear stresses, Shear flow

6.7 Shear stress distribution for various cross section

7 Strain Energy

06 Hours 07%

7.1 Elastic strain energy due to gradual loading, sudden loading, impact loading, shear and bending, Resilience

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs) :

On successful completion of the course, the student will be able to:

CO1: Classify and determine the strength parameters of materials and compute stresses & strains for structural elements due to normal, shear loads and temperature changes.

CO2: Calculate bending moment and shear force for statically determinate beams and draw the distributions.

CO3: Calculate the cross sectional moment of inertia using the parallel axis theorem.

CO4: Analyze stresses in a beam under combined loadings and also Calculate bending stress, shear stress and their distribution at any desired location along the beam elements.

CO5: Evaluate the different mechanical properties of materials and also Calculate strain energy due to different loadings.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	2	2

CO2	3	2	2	-	1	-	-	-	-	-	-	-	2	-
CO3	3	2	1	-	-	-	-	-	-	-	-	-	2	1
CO4	3	2	2	-	-	-	-	-	-	-	-	-	2	1
CO5	3	2	2	-	-	-	-	-	-	-	-	-	2	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Junarkar, S.B. & Shah, H.J., Mechanics of Structures Vol-I, Charotar Publishing House
2. Shah, H. J., Mechanics of Solids, Charotar Publishing House
3. Khurmi R. S., Strength of Materials, S. Chand Publications
4. S. S. Bhavikatti, Strength of Materials, Vikas Publishing House Pvt. Ltd
5. Hibbeler, R.C., “Mechanics of Materials”, 6th SI edition

Reference Books:

1. Beer and Johnston, Mechanics of Materials
2. Gere & Timoshenko, Mechanics of Materials, CBS Publishers & Distributors, Delhi
3. S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc.
4. R. Subramanian, Strength of Materials, Oxford University Press
5. S. S. Rattan, Strength of Materials, Tata McGraw Hill Education Pvt. Ltd.
6. R. K. Bansal, Strength of Materials, Lakshmi Publications House Pvt. Ltd.

E Books:

1. Barry Dupen, Applied Strength of Materials for Engineering Technology
2. S. Timoshenko, Strength of Materials (Part -1), D. Van Nostrand Company, Inc

Web Materials:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm>
2. <http://nptel.ac.in/video.php?subjectId=112107147>
3. <https://www.youtube.com/watch?v=GkFgysZC4Vc>
4. <http://nptel.ac.in/syllabus/112106141/>
5. <http://nptel.ac.in/courses/Webcourse-contents/IIT-Delhi/Mechanics%20Of%20Solids/index.htm>

ME – 242 Engineering Thermodynamics

3rd Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	0	3	3
Marks	100	00	100	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Fundamental Concepts & Definitions	03
2.	First-Law of Thermodynamics	06
3.	Second Law of Thermodynamics	08
4.	Entropy	10
5.	Availability and Irreversibility	06
6.	General Thermodynamic Relationships	06
7.	Properties of Gases and Mixtures	06

Total hours (Theory): 45

Total hours (Lab): 00

Total: 45

B. Detailed Syllabus:

- 1 Fundamental Concepts & Definitions 03 hours 07%
 - 1.1 Definition and scope of Thermodynamics, microscopic and macroscopic approaches.
Thermodynamic System and Control Volume, Thermodynamic Properties, path and process, quasi-static process, Thermodynamic equilibrium, Concept of Work and Heat, Zeroth law of thermodynamics.
 - 1.2 Pure Substances: Definition of a pure substance, phase of a substance, triple point and critical points, Vapour- liquid-Solid phase in pure substance with water as example. Representation of pure substance properties on P-T and P-V diagrams, P-V-T surface, Use of mollier diagram.
- 2 First Law of Thermodynamics 06 hours 16%
 - 2.1 First law of thermodynamics for a closed system undergoing (1) A cycle and (2) Change of state, Internal energy a property of the system, Perpetual motion machine of the first kind

2.2 Extension of the First law to control volume; Steady flow process, steady state-steady flow energy equation applied to nozzle, throttling process, Turbine & Compressor, Heat Exchanger, adiabatic mixing etc, Transient Flow Processes and its analysis, Filling and emptying process

3 Second Law of Thermodynamics 08 hours 18%

3.1 Identifications of directions of occurrences of natural processes, Offshoot of II law from the I, Kelvin-Planck and Clausius's statement of Second law of Thermodynamic; Equivalence of the two statements, Carnot cycle analysis.

3.2 Reversibility and irreversibility- Conditions of reversibility with examples, Carnot Theorem. Thermodynamic temperature scale, Numericals based on above topics.

4 Entropy 10 hours 20%

4.1 Introduction, Clausius equality and inequality, the property of entropy, Entropy change for open system and reversible processes, Change in entropy of ideal gases, Numericals.

4.2 Principle of increase of entropy. Third law of thermodynamics, entropy and disorder, Applications of entropy principle and Numericals.

5 Availability and Irreversibility 06 hours 12%

5.1 Exergy concept, Available energy referred to cycle, nonflow and steady flow process, irreversibility, Second law efficiency, Numericals.

6 General Thermodynamic Relationships 06 hours 13%

6.1 Introduction, Maxwell's equation, T-ds equations, Difference in heat capacities, Ratio of heat capacities

6.2 Helmholtz and Gibbs function, Internal energy relations, Clausius-Clapeyron equation, Joule-Thomson Co-efficient

7 Properties of Gases and Mixtures 06 hours 14%

7.1 Introduction to perfect, ideal and real gases, equation of state, Van der Waal's Equation of state, Van der Waal's constants in terms of critical properties

7.2 law of corresponding states, compressibility factor; compressibility chart, and other equations of state

7.3 Gas Vapor Mixtures: Mixture of Ideal Gas, mixture of perfect gases, Dalton's Law of Partial Pressure, Amagat's Law. Concept of Variable Specific heat

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.

- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the basic concepts of thermodynamics such as continuum, system, control volume, thermodynamic properties, thermodynamic equilibrium, work and heat.

CO2: Apply first law of thermodynamics to close and steady flow system.

CO3: Apply Second Law of Thermodynamics and entropy concepts to various thermodynamic systems.

CO4: Evaluate availability and irreversibility.

CO5: Derive relationship between measurable and non-measurable thermodynamic properties.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	-	-	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	-	-	-	-	-	-	-	-	1	-
CO4	3	2	1	-	-	1	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Recommended Study Material:

Text Books:

1. R. Yadav, “Fundamentals of Engineering Thermodynamics”, Central Pub House
2. P. K. Nag, “Engineering Thermodynamics”, Tata Mcgrraw-Hill Publications

Reference books:

1. Yunus a cengel, “Thermodynamics an engineering approach”, Tata Mc GRAW Hill
2. Moren M J, “Fundamentals of Engineering Thermodynamics”, John Wiley & Sons
3. Somasundaram S. L., “Engineering Thermodynamics”, New Age Pub.
4. Nagaraj P.B., “Basic Thermodynamics”, New Age Pub.
5. Arora C P, “Thermodynamics”, Tata Mc Graw Hill
6. Rao Y V C, “Theory and Problems of Thermodynamics”, Wiley Eastern Ltd

Reading Materials, web materials with full citations:

1. Steam Tables

2. Compressibility Chart
3. http://nptel.iitm.ac.in/courses/Webcourse-contents/IISc-BANG/Basic%20Thermodynamics/New_index1.html Web site: <http://www.wikipedia.org>

Other materials

1. International Journal of Thermal Sciences (Elsevier Publication)
2. Mechanical Engineering (IE)
3. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna

ME – 253 MECHANISMS AND MACHINES

3rd Semester and 2nd Year

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Machines and Mechanism	10
2.	Synthesis and Kinematic Analysis of Mechanism	14
3.	Static and dynamic analysis of mechanism	10
4.	Cams	12
5.	Gears & gear trains	14

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

1 Machines and Mechanism 10 Hours 15%

- 1.1 Introduction, Machine and Mechanism, Rigid and resistant body, Kinematic pair
- 1.2 Types of motion, Classification of Kinematic pairs, Degrees of freedom (mobility)
- 1.3 Kinematic chain, Mechanism, Kinematic analysis of planar mechanisms
- 1.4 Mobility analysis and range of movement, Grubler's criterion
- 1.5 Kutzbach's criterion, Grashof's law, Inversions of Four-bar chain and slider crank mechanism
- 1.6 Exact straight line mechanism of peaucelliers, Hart and scott- Russel, Approximate straight line mechanisms of Grasshopper, Watt and Roberts
- 1.7 Pantograph, steering gear mechanisms, Condition for correct steering, Davis and Ackermann steering gears
- 1.8 Linkage: Hooke's joint, double Hook's joint, dynamics of Hook's joint
- 1.9 Oldham's coupling, Intermittent motion mechanism

2 Synthesis and Kinematic Analysis of Mechanism 14Hours 25%

- 2.1 Introduction, Absolute and Relative Motions, Displacement
- 2.2 Introduction, Absolute and Relative Motions, Displacement by Graphical Method (Relative velocity and acceleration)
- 2.3 Coriolis Acceleration. Acceleration analysis, Klein's Construction, Instantaneous Centre of Velocity
- 2.4 Kennedy Theorem, Angular velocity ratio theorem, Centrode: Space centrode, Body centrode
- 2.5 Dimensional Synthesis of mechanism: Definitions of Type, Number and Dimensional Synthesis, Definitions of Motion, Path and Function generation, precision position, Chebychev spacing, structural error, Freudenstein's equation, two and three position synthesis (function generation only) of four bar and slider crank mechanisms by graphical and analytical methods

3 Static and dynamic analysis of mechanism 10 Hours 15%

- 3.1 Static Force Analysis: Introduction, Basic concept of equilibrium, free body diagram, Static force analysis in various types of mechanisms by Graphical Method
- 3.2 Dynamic Force Analysis: D'Alemberts principle, inertia force, Combined static and inertia force analysis of different mechanisms, Turning – moment diagrams, fluctuation of energy, flywheels

4 Cams 12 Hours 20%

- 4.1 Definitions of cam and followers their uses, Types of Cams, Types of Followers and their motion, Cam Terminology
- 4.2 Displacement Diagrams, Motion of the Followers, Analysis of motion of followers

5 Gears & gear trains 14 Hours 25%

- 5.1 Introduction, Classification of Gears, Gear Terminology, Law of Gearing, Velocity of Sliding
- 5.2 Forms of teeth, Cycloidal Profile Teeth, Involute profile teeth, Path of Contact, Arc of the contact
- 5.3 Numbers of pairs teeth in contact, Interference in involutes gears, Minimum Number of teeth
- 5.4 Interference between Rack and Pinion, Under cutting, Efficiency of Helical, Spiral, Worm, Worm Gear and Bevel Gears
- 5.5 Simple, Compound, Reverted, Epicyclic gear train, Analysis of Epicyclic gear train

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted

as a part of continuous internal theory evaluation.

- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Identify the different mechanism and its use in machine, and calculate the degree of freedom of different mechanisms.

CO2: Synthesize the planer mechanism using vector and algebraic methods Draw different Cam profile.

CO3: Determine the velocity and acceleration of different mechanism

CO4: Analyze the action of static and dynamic forces on the mechanisms.

CO5: Draw different types cam profile.

CO6: Explain working of gear mechanism and calculate the rpm and torque for different types of gear trains.

Course Articulation Matrix:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2
CO1	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	03	02	-	-	-	-	-	-	-	-	-	-	-	-
CO 4	03	01	-	-	-	-	-	-	-	-	-	-	-	-
CO5	02	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	03	02	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ambekar A. G, “Mechanism and Machine Theory”, Jain Brothers
2. Ratan S. S., “Theory of Machines”, Tata Mc Graw–Hill publications, New Delhi
3. Ghosh Amitabh, “Theory of Mechanisms and Machines”, East West Press

Reference books:

1. Rao J. S. and Dukkupati R. V., “Mechanisms and theory Machines theory”, Wiley Eastern Ltd.
2. Shigley J. E. and Uicker J. J., “Theory of Machines and Mechanisms”, Oxford University Press.

3. Bevan Thomas, “Theory of Machines” Pearson Education India.
4. Sharma C. S. and Purohit Kamlesh, “Theory of Mechanisms and Machines”, PHI.
5. Singh Sadhu, “Theory of Machines”, 2nd Ed. Pearson Education
6. Uicker John J, “Theory of Machines and Mechanism”, Oxford University Press

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-Delhi/Kinematics%20of%20Machine/index.htm>
2. <http://nptel.iitm.ac.in/video.php?courseId=1018>

Programming Languages & Software's: C, C++, MATLAB , Pro Engineer

Other materials

1. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>
2. Mechanical Engineering (IE)

ME – 251 CASTING & JOINING PROCESSES

3rd Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Manufacturing Processes	02
2.	Melting Practices	04
3.	Metal Casting Processes	16
4.	Metal Joining Processes	19
5.	Advance Welding Processes	04

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- | | | | |
|-----|--|----------|-----|
| 1 | Manufacturing Processes | 02 Hours | 4% |
| 1.1 | Basic Introduction, Importance of Manufacturing | | |
| 1.2 | Classification and Selection of Manufacturing Processes | | |
| 2 | Melting Practices | 04 Hours | 9% |
| 2.1 | Cupola: Charge calculations, construction | | |
| 2.2 | Other Furnaces : working of induction furnace, crucible furnace, and reverberate furnace | | |
| 3 | Metal Casting Processes | 16 Hours | 36% |
| 3.1 | Metal Casting, Sand Casting, Patterns | | |
| 3.2 | Cores, Gating & Riser Systems, Casting Defects & Inspection | | |
| 3.3 | Centrifugal Casting, Carbon dioxide molding process | | |
| 3.4 | Investment casting, Continuous casting | | |
| 3.5 | Shell molding, Die casting | | |
| 3.6 | Advanced technologies in casting | | |

4 Metal Joining Processes

19 Hours 42%

- 4.1 Classification of Welding, Soldering and Brazing
- 4.2 Gas Welding, Fuel Gases, Oxy–Acetylene Welding Equipment and techniques, Gas Cutting Processes
- 4.3 Types of Arc Welding, Principle of Arc, Arc Welding Equipment
- 4.4 ISI electrode classification, current and voltage selection for electrodes
- 4.5 Various arc welding processes
- 4.6 General principle of heat generation in resistance welding, Process details and working principle of different resistance welding processes.

5 Advanced Welding Processes

04 Hours 9%

- 5.1 Modern welding processes

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi–media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Categorize various manufacturing processes and choose appropriate manufacturing processes.

CO2: Express different types of melting furnaces.

CO3: Determine different casting processes and analyze casting defects.

CO4: Demonstrate mold making process and summarize the properties of molding sand.

CO5: Determine different welding processes and analyze the welding defects by penetration test.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	-	-	-	-	-	-	-	-	-	-	-	1	-

CO3	2	1	-	-	1	-	-	-	-	-	-	1	-	-
CO4	2	-	-	-	-	-	-	-	2	-	-	1	1	-
CO5	2	2	-	-	-	1	-	-	-	-	-	1	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Rao P. N., “Manufacturing Technology – Foundry, Forming & Welding”, Vol. 2, 4th Edition, Mc Graw Hill Education (India)
2. Ghosh A. and Mallik A. K., “Manufacturing Science”, 2nd edition, East West Press-2010
3. Kalpakjian S. and Schmid S.R., “Manufacturing Engineering and Technology”, 7th Edition, Pearson-2013
4. Sharma P.C., “A Textbook of Production Technology (Manufacturing Processes)”, 8th Edition, S. Chand Publishing

Reference books:

1. Kumar Rajeev, “Manufacturing Processes”- Prentice Hall India Learning Private Limited -2014
2. Groover M. P., “Fundamentals of Modern Manufacturing”, 5th edition, Wiley-2012
3. Lindberg Roy A, "Processes and materials of manufacture", Fourth edition PHI, 1990
4. Schey J., “Introduction to Manufacturing Processes”, Tata McGraw Hill -1999
5. Parmar R.S., “Welding Processes and Technology “, Khanna Publishers
6. Jain P.L., “Principles of Foundry Technology”, 5th Edition, Mc Graw Hill Education (India)
7. Ravi B., “Metal Casting: Computer Aided Design and Analysis”, PHI Publication

Reading Materials, web materials with full citations:

1. <http://nptel.iitm.ac.in/courses/Webcourse-contents/IIT-ROORKEE/MANUFACTURING-PROCESSES/index.htm>
2. Foundry Informatics Centre, New Delhi, <http://www.foundryinfo-india.org>

Other materials

1. Journal of Material Processing Technology (Elsevier Publication)
2. Production Engineering (IE) <http://www.ipcourseline.org>
3. Sadhana-Indian Academy of Sciences, <http://www.ias.ac.in/sadhana>
4. E-Foundry, <http://efoundry.iitb.ac.in>
5. International journal of metal casting (Springer Publication)

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

3rd Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	100		100	

A. Objective of the Course:

To facilitate learners to:

- gain familiarity with the mechanics of creativity and problem solving
- develop an attitude for innovation
- develop creative thinking skills using cone of learning components leading to understanding of strategies of creativity, problem solving and innovation
- explore applications of the concepts of creativity and problem solvings skills in personal, social, academic, and profession life.

B. Outline of the Course:

Module No.	Title/Topic	Classroom Contact Hours
1	Introduction to Creativity, Problem Solving and Innovation • Definitions of Creativity and Innovation • Need for Problem Solving and Innovation • Scope of Creativity in various Domains • Types and Styles of Thinking • Strategies to develop Creativity, Problem Solving and Innovation skills	06
2	Questioning, Learning and Visualization • <i>Strategy and Methods of Questioning</i> • <i>Asking the Right Questions</i> • <i>Strategy of Learning and its Importance</i> • <i>Sources and Methods of Learning</i> • <i>Purpose and Value of Creativity Education in real life</i> • <i>Visualization strategies - Making thoughts Visible</i> • <i>Mind Mapping and Visualizing Thinking</i>	06
3	Creative Thinking and Problem Solving • <i>Creative Thinking and its need</i> • <i>Strategy of Thinking Fluency</i> • <i>Generating all Possibilities</i> • <i>SCAMPER Technique</i>	06

	• <i>Divergent Vs Convergent Thinking</i> • <i>Lateral Vs Vertical Thinking</i> • <i>Fusion of Ideas for Problem Solving</i> • <i>Applying strategies for Problem Solving</i>	
4	Logic, Language and Reasoning • <i>Basic Concepts of Logic</i> • <i>Statement Vs Sentence</i> • <i>Premises Vs Conclusion</i> • <i>Concept of an Argument</i> • <i>Functions of Language: Informative, Expressive and Directive</i> • <i>Inductive Vs Deductive Reasoning</i> • <i>Critical Thinking & Creativity</i> • <i>Moral Reasoning</i>	06
5	Contemporary Issues and Practices in Creativity and Problem Solving • <i>Cognitive Research Trust Thinking for Creatively Solving Problems</i> • <i>Case Study on Contemporary Issues and Practices in Creativity and Problem Solving</i>	06
Total		30

C. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by Slides Presentations, Reading Material, Discussions, Case Studies, Puzzles, Ted Talks, Videos, Task-Based Learning, Projects, Assignments and various Individual and Interpersonal activities like, Critical reading, Group work, Independent and Collaborative Research, Presentations, etc.

D. Evaluation

There will be no end semester university examinations. Students will be evaluated continuously in the form of internal as well as external evaluation. The evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation. The concerned teacher shall evaluate students distribute the marks (out of 30 as Internal and out of 70 as External) and submit them. There may also be a Computer Based Test at the end.

Evaluation Scheme

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Attendance	100%	--	20
2	Individual Activity Participation	As stipulated by the		20

3	Group Activity Participation	Resource Person(s) in the Training	20
4	Presentation		30
5	Feedback on Improvement		10
Total			100

E. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Demonstrate creativity in their day to day activities and academic output.

CO2: Solve personal, social and professional problems with a positive and an objective mindset.

CO3: Think creatively and work towards problem solving in a strategic way.

CO4: Initiate new and innovative practices in their chosen field of profession.

CO5: Give logical ideas, opinions, and solutions to problems.

CO6: Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	1	1	-	-	-	2	-	1	1	-
CO2	-	3	2	1	-	-	-	1	1	1	-	1	-	1
CO3	-	2	2	1	-	-	-	-	-	-	-	-	-	-
CO4	1	-	1	-	-	-	2	-	-	-	-	-	1	-
CO5	-	1	-	1	-	-	-	-	1	3	-	1	-	1
CO6	-	-	-	2	-	-	-	-	-	-	2	-	1	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

F. Books/Reference Books / Reading

Text Books

1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity, Jossey-Bass Publication 2013
2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

Reference Books

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno's Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map, Thorsons Publication 2002
5. Scott Berkum, The Myths of Innovation, Expanded and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992

8. Paul Sloane, Des MacHale & M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book, Sterling Publication 2002

Additional Readings

1. Keith Sawyer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman, The Art of Innovation, Profile Publication 2001
5. Roger Von Oech, A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer, Imagine How Creativity Works, Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication 1994
9. Scott G Isaksen, K Brain Doval, Donald J Treffinger, Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, Scott G Isaksen, K Brainstead Dorval Creative Problem Solving An Introduction, Prufrock Press 2006
11. H Scott Fogler & Steven E. LeBlance, Strategies for Creative Problem Solving, Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds, Basic Books Publication 1993
14. Mihaly Csikszentmihalyi, Creativity-Flow and Psychology of Discovery and Invention, Harper Publication 1996
15. Martin Gerdner, W. H., Aha! Insight, Freeman Publication 1978
16. Paul Sloane, Test Your Lateral Thinking IQ, Sterling Publication 1994
17. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles, Sterling Publication 1996

Articles / Videos / Other Suggested Materials

- Internet Search based May TED talks and other sources for videos, slide shares, problems, Etc.

B. Tech. (Mechanical Engineering) Programme

SYLLABI (Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

ME – 252 MACHINING & FORMING PROCESSES

4th Semester and 2nd Year

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/Week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No	Title of Unit	Minimum number of Hours
1.	Metal Cutting Lathes	08
2.	Drilling and Boring Machines	03
3.	Milling Machines	04
4.	Planers, Shapers and Slotters	03
5.	Sawing and Broaching Machines	02
6.	Grinding Machines and Abrasives	03
7.	Fundamentals of Metal Forming	07
8.	Rolling	03
9.	Forging	04
10.	Extrusion	03
11.	Drawing Operations	02
12.	Press Working and Dies	03

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- 1 **Metal Cutting Lathes** 08 Hours 18%
- 1.1 Engine Lathes, Construction, Arrangement and Principle, Units of engine lathes, Type and size range of engine lathes.
- 1.2 Operations carried on engine lathe, Attachment extending the processing capacities of engine lathes.
- 1.3 Description of other types of lathes, Plain turning lathes, Facing lathes, Multiple tool lathes.

1.4	Simple purpose lathes, Turret lathes, Horizontal and Vertical lathes.		
2	Drilling and Boring Machines	03 Hours	07%
2.1	Purpose and field of application of drilling and boring machines.		
2.2	Upright drill processes, radial drills, Horizontal and Precision Boring Machines.		
3	Milling Machines	04 Hours	08%
3.1	Purpose and types of milling machines, general purpose milling machines.		
3.2	Different types of milling operations.		
3.3	Milling cutters, attachments extending the processing capabilities of general purpose milling machines.		
4	Planers, Shapers and Slotters	03 Hours	07%
4.1	Classification, Attachments extending the processing capacities of each.		
5	Sawing and Broaching Machines	02 Hours	04%
5.1	Metal sawing – classification; Reciprocating sawing machines, Circular sawing machines, Band sawing machines.		
5.2	Types of broaching machines, Advantages and Limitations of Broaching.		
6	Grinding Machines and Abrasives	03 Hours	07%
6.1	Classifications of grinding machines, Cylindrical grinders, Internal grinders, Surface grinders, Tool and Cutter grinders.		
6.2	Surface finishing, Abrasives, Manufacture of grinding wheels.		
7	Fundamentals of Metal Forming Processes	07 Hours	16%
7.1	Introduction, Classification of Forming Processes, Mechanics of Metal Working.		
7.2	Various Temperatures in Metal Working, Cold and Hot Working. Formability, Strain Rate Effects on metal forming, Effects of Metallurgical Structure on Metal Forming, Hydro Static Pressure, Residual Stresses.		
8	Rolling	03 Hours	07%
8.1	Introduction and classification of Rolling processes, Principles of Metal Rolling, Simplified Analysis of Rolling Load, Various Rolling Parameters.		

8.2 Defects in rolled products and remedies of it.

9 Forging 04 Hours 08%

9.1 Introduction and classification of Forging Processes, Various Forging Operations, Forging Die Materials and Lubrication, Forge ability, Forging Defects and remedies.

10 Extrusion 03 Hours 07%

10.1 Introduction and classification of Extrusion Processes, Various Extrusion Operations, Metal Deformation and Forces in Extrusion.

10.2 Materials and Lubrication considerations in Extrusion Process, Extrusion Defects, Extrusion of Tubing, Production of Seamless Pipe and Tubing.

11 Drawing Operations 02 Hours 04%

11.1 Drawing of Rods, Wires and Tubes, Sizing.

12 Press Working and Dies 03 Hours 07%

12.1 Types of presses drive and feed mechanisms, press tools.

12.2 Various press working operations and its parameters, Elements of press, Various Metal Forming Operations.

12.3 Stock strip layout, study of sheet metal nesting software.

C. Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Understand working of lathe, shaper and planer, drilling, milling and grinding machines.

CO2: Comprehend speed and feed mechanisms of machine tools.

CO3: Study and practice on machine tools and their operations..

CO4: Estimate machining times for machining operations on machine tools.

CO5: Gain knowledge of classification of metal forming processes, plastic deformation, flow stress, formability, strain, effect of temperature, strain rate and metallurgical structure on metal working, friction and lubrication, deformation zone geometry, workability, and residual stresses.

CO6: Understand metal forming processes like forging, drawing, rolling, extrusion, and deep drawing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	1	1	-
CO2	3	1	-	-	-	-	-	-	-	-	-	1	1	-
CO3	1	2	-	1	-	-	-	-	-	-	-	3	2	2
CO4	2	3	-	1	-	-	-	-	-	-	-	1	2	1
CO5	3	1	-	1	-	-	-	-	-	-	-	1	1	-
CO6	2	2	-	-	-	-	-	-	-	-	-	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

Rao P. N., “Manufacturing Technology Vol. 2 – Metal Cutting and Machine Tools”, Tata McGraw Hill, 3rd Edition, 2013.

Rao P. N., “Manufacturing Technology – Foundry, Forming and Welding”, Tata McGraw Hill, 4th Edition, 2013.

Jain.R.K, “Production Technology: Manufacturing Processes, Technology and Automation”, Khanna Publishers, 17th Edition, 2011.

Reference Books:

Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. – I”, Media Promoters, 2008.

Hajara Choudhury S. K., “Elements of Work–Shop Technology Vol. –II”, Media Promoters, 2010.

Kalpajakjian S., “Manufacturing, Engineering and Technology”, Pearson Education Canada, 7th Edition, 2013.

Chapman W. A., “Workshop Technology Vol. I and II and III”, CBS publishers & distributors pvt. ltd.

Web Material:

<http://nptel.iitm.ac.in/courses.php?branch=Mechanical>

Other Material:

<http://www.sciencedirect.com/science/journal/09240136>

<http://www.ieindia.org/publish/pr/pr.htm>

<http://www.ieindia.info/public.asp/me/me.htm>

<http://www.ias.ac.in/sadhana/>

ME - 246 Material Engineering & Metallurgy

4th Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Introduction to material Engineering and metallurgy	03
2.	Crystal Structure and Imperfection	04
3.	Phase Diagrams and Iron Carbon System	04
4.	Cast Iron and wrought iron	04
5.	Alloy Steel	04
6.	Heat Treatment of Steels	12
7.	Non-ferrous alloys	04
8.	Powder Metallurgy	06
9.	Corrosion and degradation of metals	05
10.	Materials testing and inspection	08
11.	Composite Materials	06

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

- | | | | |
|-----|---|----------|----|
| 1 | Introduction to Material Engineering and Metallurgy | 03 Hours | 5% |
| 1.1 | Classification of Engineering Materials. | | |
| 1.2 | Engineering requirements of materials. | | |
| 1.3 | Properties of engineering materials. | | |
| 1.4 | Criteria for selection of materials for engineering applications. | | |
| 1.5 | Concept of Plastics Materials, Classification, Properties and Application of plastics. | | |
| 1.6 | Concept of Electronic Materials, Classification, Properties and Application of Electronics Materials. | | |
| 1.7 | Types of Metallurgical Processes. | | |
| 2 | Crystal Geometry and Imperfections | 04 Hours | 7% |

2.1	Crystal systems, crystallographic points, directions and planes.		
2.2	X ray diffraction and Bragg's law.		
2.3	Imperfection in solids – point, line, and surface defects, Dislocation moments.		
3	Phase Diagrams and Iron Carbon System	04 Hours	7%
3.1	Gibbs phase rule and phase diagrams Type I, II, III.		
3.2	Iron carbon diagram.		
3.3	Structure Properties Relationship.		
4	Cast Iron and wrought iron	04 Hours	7%
4.1	Grades of C.I.		
4.2	Alloyed Cast Iron, Malleable Iron and S. G. Iron.		
4.3	Wrought Iron Properties and uses.		
5	Alloy Steel	04 Hours	7%
5.1	Classification of Steels.		
5.2	Properties and uses.		
5.3	Effects of different alloying metals.		
6	Heat Treatment of Steels	12 Hours	18%
6.1	Study of Heat-Treatment processes and TTT diagram.		
6.2	Normalizing, Annealing, Spheroidizing.		
6.3	Hardening, tempering, austempering.		
6.4	Martempering, case-hardening, nitriding, cyaniding.		
6.5	Induction hardening, flame-hardening, ageing.		
6.6	Application of above processes in mechanical components such as gears, bearing, turbine blades, crankshafts, pistons, cutting tool materials.		
7	Non-ferrous alloys	04 Hours	6%
7.1	Extraction of copper, aluminum and nickel.		
7.2	Alloys of copper, aluminum, Magnesium, lead, tin, zinc, nickel, titanium, bearing alloys.		
8	Powder Metallurgy	06 Hours	10%
8.1	Application, advantages and limitation of Powder metallurgy.		
8.2	Production of powder.		
8.3	Compacting, Sintering.		
8.4	Equipment and process capability.		
9	Corrosion and degradation of metals	05 Hours	7%
9.1	Meaning, causes and nature of corrosion		

- 9.2 Intergranular corrosion (IGC), Hydrogen embrittlement
- 9.3 Measures of counteracting corrosion
- 9.4 Metal coatings, Organic coatings, Lining and cladding, Use of Corrosion inhibitors, cathodic protection against corrosion.

10 Materials testing and Inspection **08 Hours 18%**

- 10.1 Introduction to Non-destructive testing.
- 10.2 Radiography Testing.
- 10.3 Dye Penetration Testing, Magnetic Particle Testing.
- 10.4 Ultrasonic Testing.
- 10.5 Jominy end quench test.
- 10.6 Macro-examination, Spark Test, Macro-etching, Fracture.
- 10.7 Microscopic examinations.
- 10.8 Electron Microscopy, Magnetic Testing.
- 10.9 Chemical analysis of steel and Iron.

11 Composite Materials **06 Hours 08%**

- 11.1 Types and application of composites.
- 11.2 Particle reinforce composites, Influence of fiber orientation and concentration.
- 11.3 Fiber phase and the matrix phase, Polymer matrix composite.
- 11.4 Metal matrix composite, ceramic matrix composite and carbon-carbon composites.

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- In the lectures the discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Explain the crystal structure and classification of materials and Explore various extraction processes to get the different form of materials.

CO2: Interpret the phase diagrams of materials.

CO3: Classify cast irons, wrought iron and explain their applications.

CO4: Compare different heat treatment processes to achieve desired properties of metals and alloys.

CO5: Understand the concept of materials testing and inspection.

CO6: Develop concept of corrosion and its prevention.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	2	-	-	-	2	-	-	-	-	2	-	-
CO2	3	2	2	2	2	-	-	-	-	-	-	-	1	-
CO3	3	3	-	3	-	-	-	-	-	-	-	2	1	-
CO4	3	2	3	2	2	2	2	-	-	-	-	-	-	-
CO5	2	-	3	3	3	-	-	-	-	-	-	2	-	-
CO6	2	2	2	2	2	-	-	-	-	-	-	2	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Callister W. D. Jr, “Materials science and engineering : An Introduction”, Edition:-2006, Wiley India, New Delhi, India
2. Narang G. B. S. and Manchanedy K., “Materials and Metallurgy”, Khanna Pub New Delhi, India
3. Swaroop D. and Saxena M. N., “Elements of Metallurgy”, Rastogi Pub, Meerut, India
4. Dharmendra K.umar and Jain S. K., “Material science and manufacturing process”, Vikas Publishing House, New Delhi, India
5. Reed R. E., “Physical Metallurgy Principles”, Cengage Learning, 2009

Reference books:

1. Raghavan V., “Metallurgy for engineers”, Prentice Hall of India
2. Avner Sidney H., “Physical Metallurgy”, Tata McGraw Hill Education
3. Khanna O. P., “Material Science - A Text Book of Material Science & Metallurgy”, Dhanpat Rai Publisher

Reading Materials, web materials with full citations:

1. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Material%20Science/New_index1.html
2. <http://ocw.mit.edu/OcwWeb/web/courses/courses/index.htm#MaterialsScienceandEngineering>

Other materials

1. Metallurgical & Materials Engineering (IE)
2. Sadhana (Engineering Sciences) www.ios.ac.in/sadhna
3. NDT & E International (Elsevier Publications)

ME – 247 FLUID MECHANICS

4th Semester and 2nd Year

Credit and Hours::

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Properties of fluid	02
2.	Fluid Statics	08
3.	Kinematics of fluid flow	08
4.	Fluid Dynamics	05
5.	Flow Measurement	04
6.	Vortex Flow	04
7.	Flow Through Pipes	04
8.	Viscous Flow	05
9.	Dimensional analysis	05

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- 1 Properties of fluid 02Hours 04%
 - 1.1 Introduction, classification of fluids
 - 1.2 Study of fluid properties
- 2 Fluid Statics 08Hours 18%
 - 2.1 Pressure at a point, center of pressure on plane and curved surface
 - 2.2 pressure measurement with various manometers with numerical problems
 - 2.3 Buoyant force, stability of submerged body and floating body.
 - 2.4 Meta centre & Meta centric height – analytical and experimental determination with problems

- | | | | |
|----------|--|----------------|------------|
| 3 | Kinematics of fluid flow | 08Hours | 18% |
| 3.1 | Description of fluid flow using Lagrangian and Eulerian method, Continuity equations for 2-D and 3-D flow in Cartesian coordinates of system | | |
| 3.2 | Classification of fluid flows, Stream line, path line, streak line, stream tube | | |
| 3.3 | Rotational and ir-rotational flow, circulation and vorticity, stream functions, velocity potential and potential flow, flow nets | | |
|
 | | | |
| 4 | Fluid Dynamics | 05Hours | 11% |
| 4.1 | Euler's equation along stream tube and in Cartesian Co-ordinates | | |
| 4.2 | Bernoulli's equation in one dimension flow and problems | | |
| 4.3 | Momentum equation for 2-D and 3-D flow along a stream line, Kinetic energy correction and momentum correction factor | | |
|
 | | | |
| 5 | Flow Measurement | 04Hours | 09% |
| 5.1 | Measurement of flow with pitot tube, venturimeter, orifice-meter, nozzle, bendmeter | | |
| 5.2 | Elementary theory of notches and weirs, flow in a curved path, pressure gradient and change of total energy across the streamlines | | |
|
 | | | |
| 6 | Vortex Flow | 04Hours | 09% |
| 6.1 | Introduction, Forced Vortex flow, Free Vortex Flow | | |
| 6.2 | Equation of Motion for Vortex Flow, Equation of Free & Forced Vortex Flow | | |
|
 | | | |
| 7 | Flow Through Pipes | 04Hours | 09% |
| 7.1 | Loss of Energy in pipes, Frictional loss in pipe flow, Chezy's and Darcy Equation for loss of Head Due to friction in pipes | | |
| 7.2 | Flow through compound, parallel and branched pipes | | |
|
 | | | |
| 8 | Viscous Flow | 05Hours | 11% |
| 8.1 | Introduction, Reynolds experiment | | |
| 8.2 | Flow between two parallel fixed plates, Couette flow, viscous flow through pipes, Hagen – Poiseuille's equation friction factor, Moody diagram, Loss of head due to friction | | |
| 8.3 | Determination of coefficient of viscosity using viscometer – Capillary tube type, rotating cylinder, falling sphere, Saybolt/ Redwood viscometer | | |
|
 | | | |
| 9 | Dimensional analysis | 05Hours | 11% |
| 9.1 | Introduction, Fundamental dimensions, dimensional homogeneity | | |
| 9.2 | Rayleigh's method and Buckingham's method | | |
| 9.3 | Dimensionless numbers and their significance, Hydraulic similitude, Type of models, Model | | |

C. Instructional Methods and Pedagogy:

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Calculate the Fluid Properties.

CO2: Solve the various fluid static as well as kinematics problems.

CO3: Apply Euler's and Bernoulli's equations for incompressible and in-viscid fluid flow.

CO4: Explain the dynamics of fluid flow and mechanics of viscous flow.

CO5: Analyze pipe flows and open-channel flows.

CO6: Perform dimensional analysis for problems in fluid mechanics.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	1	-	-	-	-	-	-	-	-	1	-
CO2	3	3	1	-	-	-	-	-	-	-	-	-	1	-
CO3	3	3	1	-	-	-	-	-	-	-	-	-	2	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	3	1	-	-	-	-	-	-	-	-	-	2	-
CO6	3	-	-	-	-	-	-	-	-	-	-	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Bansal R. K., “Fluid Mechanics”, Laxmi Publication
2. Rajput R. K., “Fluid Mechanics”, S. Chand & Co
3. Kumar D. S., “Fluid Mechanics and Fluid Power Engineering”, S K Kataria Publishers, New Delhi

4. Khurmi R. S. “Fluid Mechanics and Hydraulic Machines”, S. Chand & Co

Reference Books:

1. Subramanya K., “Theory and Applications of Fluid Mechanics”, Tata–McGraw Hill Publishing Co, 1993, New Delhi
2. White F. M. “Fluid Mechanics”, McGraw Hill, New York
3. Modi & Sheth, “Fluid mechanics & Hydraulic Machines”, Standard book house

web material:

1. <http://www.nptel.iitm.ac.in/courses/Webcourse-contents/IIT-KANPUR/FLUID-MECHANICS/ui/About-Faculty.html>

Other materials

1. Mechanical Engineering (IE)
2. Flow Measurement and Instrumentation (Elsevier Publications)
3. International Journal of Heat and Fluid Flow (Elsevier Publications)
4. Journal of Fluids and Structures (Elsevier Publications)
5. Sadhna (Engineering Sciences) www.ios.ac.in/sadhna

ME – 250 DYNAMICS OF MACHINES

4th Semester and 2nd Year

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Mechanical vibration of single degree freedom system	10
2.	Mechanical vibration of two degree and multi-degree freedom system	08
3.	Governors	08
4.	Gyroscope	08
5.	Friction	10
6.	Balancing	12
7.	Cam Dynamics	04

Total hours (Theory): 60

Total hours (Lab): 30

Total: 90

B. Detailed Syllabus:

- 1 Mechanical vibration of single degree freedom system 10Hours 17%
 - 1.1 Basic concepts of Vibration, Classification of vibration.
 - 1.2 Free vibration of undamped system.
 - 1.3 Free damped vibration with viscous damping, Coulomb damping.
 - 1.4 Response of an damped system under harmonic force , Response of an damped system under harmonic motion of the base and under rotating unbalance.
- 2 Mechanical vibration of two degree and multi-degree freedom system 08Hours 13%
 - 2.1 Free un-damped vibrations – Principal modes and natural frequencies, Co-ordinate coupling, vibration absorber.
 - 2.2 Multi-degree of Freedom Systems: Hozler's method, Matrix method

3 Governors 08Hours 13%

- 3.1 Types: Centrifugal, dead weight and spring controlled governors
- 3.2 Controlling force, effort and power of a governor
- 3.3 Sensitiveness, hunting and isochronisms in governors

4 Gyroscope 08Hours 13%

- 4.1 Gyroscopic couple, gyroscopic stabilization of ship and vehicles
- 4.2 Application of the principle of gyroscopic couple to two wheelers
- 4.3 Four wheelers, planes and ships etc.

5 Friction 10Hours 17%

- 5.1 Introduction, Kinds of friction, Laws of friction, Coefficient of friction, Surface friction
- 5.2 Friction in screws with square thread and V threads, Pivot and collar friction, Rolling Friction

6 Balancing 12Hours 21%

- 6.1 Introduction, Static and dynamic balance, Balancing of rotating masses in different planes
- 6.2 Dynamics of Reciprocating Machines with Single Slider; Unbalance in Single Cylinder Engine Mechanisms
- 6.3 Unbalance in Multicylinder Engines –In–line, V–twin and Radial Engines; Balancing Techniques, Field Balancing, balancing machines.

7 Cam Dynamics 04Hours 6%

- 7.1 Analysis of circular arc cam and tangent cam, dynamics of high speed cam systems
- 7.2 Polydyne cams, force analysis of cams, vibrations, jump, Shock
- 7.3 Spring surge criteria in high speed cams

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Derive an equation of motion and find out natural frequency of single and multi-degree freedom system, also analyze the effect of external force forced vibration and damped vibration.

CO2: Calculate the height of the governor or speed of governor.

CO3: Identify and describe the gyroscopic couple and reactive gyroscopic couple for ship, aero plane and motorcycle.

CO4: Apply theory of friction to find out the torque for the screw jack, bearing and frictional clutches.

CO5: Solve the problem of rotating and reciprocating unbalance system.

CO6: Illustrate the high speed cam and jump phenomena.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	2	2	-	-	-	-	-	-	-	3	-
CO2	3	2	1	2	-	-	-	-	-	-	-	-	2	-
CO3	3	2	-	1	-	-	-	-	-	-	-	-	3	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	3	-
CO5	3	2	2	2	1	-	-	-	-	-	-	-	3	-
CO6	3	3	-	2	-	-	-	-	-	-	-	-	2	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Ambekar A G, “Mechanism and Machine Theory”, Jain Brothers
2. Ratan S. S., “Theory of Machines”, Tata Mc Graw–Hill publications, New Delhi.
3. Ghosh Amitabha, “Theory of Mechanisms and Machines”, East West Press

Reference books:

1. Rao J. S. and Duggipati R. V. “Mechanisms and theory Machines theory”, Wiley Eastern Ltd.
2. Shigley J. E and Uicker J. J., “Theory of Machines and Mechanisms”, Oxford University Press.
3. Norton R. L, “Kinematics & Dynamics of Machinery “,
4. Bevan Thomas, “Theory of Machines” Tata McGraw-Hill publications, New Delhi.
5. Ballaney P L, “Theory of Machine & Mechanism”
6. Singh Sadhu, “Theory of Machines”, 2nd Ed. Pearson Education
7. Uicker John J, “Theory of Machines and Mechanism”, Oxford University Press

Web Materials

1. <http://www.nptel.iitm.ac.in/>

Other materials

1. Programming Languages & software's: C, C++, MATLAB , Pro Engineer
2. SADHNA (Engineering Science): <http://www.ias.ac.in/sadhana/>

MA – 249 NUMERICAL METHODS

4th Semester and 2nd Year

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1.	Errors	05
2.	Approximate Solutions of Nonlinear Equations and System of Linear Equations	08
3.	Interpolation	12
4.	Numerical Integration and Differentiation	08
5.	Numerical solution of Ordinary and Partial Differential Equation	12

Total hours (Theory): 45

Total hours (Lab): 30

Total: 75

B. Detailed Syllabus:

- | | | | |
|----------|--|-----------------|------------|
| 1 | Errors | 05 Hours | 12% |
| 1.1 | Exact and Approximate Numbers and Significant Figures | | |
| 1.2 | Rules of Rounding of Numbers, Correct Digit | | |
| 1.3 | Types of Errors: Absolute, Relative and Percentage | | |
| 1.4 | Rounding Error, Inherent Error and Truncation Error | | |
| 1.5 | Propagation of Errors | | |
| 2 | Approximate solutions of nonlinear equations and System of linear equations | 08 Hours | 18% |
| 2.1 | Bisection method, False Position Method, Method of Iteration | | |
| 2.2 | Newton-Raphson method for single variable and their convergence | | |
| 2.3 | Gauss Jordan method | | |
| 2.4 | Gauss Seidel method and Gauss Jacobi method | | |

3	Interpolation	12 Hours	26%
3.1	Finite differences and associated operators		
3.2	Newton's Interpolation formulas		
3.3	Lagrange interpolation, Hermite interpolation		
3.4	Error estimates of these formulae		
3.5	Piecewise Interpolation : Linear Spline, Cubic Spline		
4	Numerical Integration and Differentiation	08 Hours	18%
4.1	Newton-Cotes Quadrature formula		
4.2	Trapezoidal rule with error		
4.3	Simpson's 1/3 rule and Simpson's 3/8 rule with respective errors		
4.4	Romberg Integration and Gaussian Quadrature		
4.5	Numerical Differentiation		
5	Numerical solution of Ordinary and Partial differential equation	12 Hours.	26 %
5.1	Overview of ordinary differential equations and partial differential equations		
5.2	Euler method and Modified Euler method		
5.3	Runge -Kutta 2 nd and 4 th order methods		
5.4	Finite difference approximations to derivatives		
5.5	Solution of Laplace Equation by Gauss Jacobi and Gauss Seidel Method		
5.6	Solution of Heat Equation in One dimension		
5.7	Solution of Wave Equation in One dimension		

C. Instructional Methods and Pedagogy

- At the starting of the course, delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, etc.
- Attendance is compulsory in lectures and laboratory.
- Internal exams/Unit tests/Surprise tests/Quizzes/Seminar/Assignments etc. will be conducted as a part of continuous internal theory evaluation.
- The course includes a laboratory, where students will get opportunities to build appreciation for the concepts being taught in lectures using software.
- Experiments/tutorials related to course content will be carried out in the laboratory.
- In the lectures and laboratory discipline and behavior will be observed strictly.

D. Course Outcomes (COs):

On successful completion of the course, the student will be able to:

CO1: Employ the knowledge of numerical methods to Engineering and real world problems.

CO2: Determine approximate solution of non-linear equations and system of linear equations and examine the error using numerical methods.

CO3: Use numerical techniques for differentiation and definite integration.

CO4: Interpolate or extrapolate approximate value of dependent variable for any value of independent variable, employing only finitely many tabulated values.

CO5: Find the solution of ordinary and partial differential equation numerically.

CO6: Construct the algorithm and perform the various commands in programming language.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	POS01	POS02
CO1	3	3	1	2	-	-	-	-	-	-	-	1	1	-
CO2	3	3	1	2	-	-	-	-	-	-	-	-	1	1
CO3	3	3	1	2	-	-	-	-	-	-	-	-	1	-
CO4	3	3	1	2	-	-	-	-	-	-	-	-	1	1
CO5	3	3	1	2	-	-	-	-	-	-	-	-	1	1
CO6	2	2	-	2	1	-	-	-	-	-	-	1	1	1

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation “-”

E. Recommended Study Material:

Text Books:

1. Sastry, S. S. Introductory methods of numerical analysis. PHI Learning Pvt. Ltd., 2012.
2. Chapra, Steven C., and Raymond P. Canale. Numerical methods for engineers. Vol. 2. New York: McGraw-Hill, 2012.

Reference books:

1. Grewal, B. S., & Grewal, J. S. Numerical Methods in Engineering & Science: With Programs in C, C++ & MATLAB. Khanna, 2014.
2. Jain, M. K., Iyengar, S. R. K. and Jain, R. K., "Numerical Methods for Scientific and Engineering Computation", New Age Pvt. Pub, New Delhi, 2000.
3. K. E. Atkinson, Introduction to Numerical Analysis, 2nd Edn, John Wiley, 1989.

Reading Materials, web materials with full citations:

1. <http://numericalmethods.eng.usf.edu>, <http://nptel.ac.in>
2. <http://mathworld.wolfram.com/>

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester IV
HSIII.03: Human Values and Ethics

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	2
Marks	--	100	--	100	

Pre-requisite:

- Basic awareness of Values and Ethics

Objectives of the Course:

To facilitate learners to:

- Develop a familiarity with the Fundamental Human Values and Ethics
- Understanding Values and Ethics as per Ancient Indian Tradition
- Apply Values and Ethics in their professional, social and personal spectrum
- Understand and apply Constitutional Values and Fundamental Duties
- Explore global issues and propose solutions with reference to values and Ethics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Ethics and Values in Ancient Indian Tradition	08
3.	Types of Values and Applied Ethics	08
4.	Constitutional Values and Fundamental Duties	05
5.	Global Issues of Values and Ethics in Contemporary Society	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1. **Introduction to Values and Ethics** 5 Hours 17%
Need, Relevance and Significance of Values, Concept and Meaning of Values and Ethics
2. **Ethics and Values in Ancient Indian Tradition** 8 Hours 26%
Lessons from Jatakkatha
Lessons from Panchtantra Tales
Ethics to be followed from Ancient Hinduism, Jainism and Buddhism
Selected Anecdotes from ancient mythology and Moral Learnings
3. **Types of Values and Applied Ethics** 8 Hours 26%
Personal Values, Social, Civic & Democratic Value, Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics
4. **Constitutional Values and Fundamental Duties** 5 Hours 17%
Basics of Indian Constitution
Constitutional Values: Justice, Liberty, Equality and Fraternity
Fundamental Rights
Fundamental Duties
5. **Global Issues of Values and Ethics in Contemporary Society** 4 Hours 14%
Case Studies, Presentations, Projects

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the significance of values and ethics
CO2	Understand concept of values and ethics as per Ancient Indian Tradition
CO3	Understand and apply various types of Values and Applied Ethics
CO4	Develop awareness and initiate implementation of Constitutional Values and Fundamental Duties
CO5	Get aware of the global issues and scope and application of values and ethics in problem solving

Evaluation Scheme

The evaluation scheme for the course will comprise the following components:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Reference book:

1. Human Values and Ethics in Workplace, United Nations Settlement Program, 2006.
(http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
2. Ethics for Everyone, Arthur Dorbin, 2009.
(<http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf>) .
3. Values and Ethics for 21st Century, BBVA.
(https://www.bbvaopenmind.com/wp-content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

❖ Web material:

- www.ethics.org

Note: The subtopics mentioned in the modules 2 and 3 are comprehensive. The faculty member can choose any 2 topics from module and 5 topics from module 3.

Charotar University of Science and Technology (CHARUSAT)
Faculty of Humanities (FoH)
Semester-IV

HS113 A: Indian Knowledge System

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	02	--	30	02
Marks	--	100	--	100	

Pre-requisite:

- --

Objectives of the Course:

- to develop a foundational understanding of the Indian Knowledge System, its origins, and its significance in shaping Indian culture and society.
- to analyze the Vedic texts, their content, and their cultural and philosophical implications.
- to investigate major philosophical schools in India, including their core tenets and historical context.
- to understand the continuity and evolution of knowledge and wisdom in India across different time periods.
- to gain insight into the role of linguistics and languages in preserving and transmitting knowledge within the Indian context.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	An overview of Indian Knowledge System (IKS)	04
2.	The Vedic Corpus	06
3.	Indian Philosophical Systems	04
4.	Wisdom through the Ages	04
5.	Linguistics	06
6.	The knowledge triangle	06
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1	An overview of Indian Knowledge System (IKS):	04
	▪ Importance of Ancient Knowledge ▪ Definition of IKS ▪ Classification framework of IKS ▪ Unique aspects of IKS	
2	The Vedic Corpus:	06
	▪ Vedas and Vedangas ▪ Distinctive features of Vedic Life	
3	Indian Philosophical Systems:	04
	▪ Introduction to Indian Philosophies ▪ Different schools of philosophy	
4	Wisdom through the Ages:	04
	▪ Puranas ▪ Itihasas ▪ Niti shastras ▪ Subhasitas	
5	Linguistics:	06
	▪ Components of a language ▪ Paṇini's work on Sanskrit grammar ▪ Phonetics in Sanskrit ▪ The role of Sanskrit in natural language processing	
6	The knowledge triangle:	06
	▪ Prameya ▪ Pramaṇa ▪ Saṃsaya ▪ Framework for establishing Valid Knowledge ▪ Potential fallacies in the reasoning process	

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	describe the fundamental concepts and components of the Indian Knowledge System, including its historical development.
CO2	analyze select Vedic texts, extracting their philosophical and cultural significance.
CO3	differentiate between major Indian philosophical systems, such as Vedanta, Nyaya, and Sankhya, and comprehend their key principles.
CO4	trace the evolution of wisdom and knowledge in India from ancient times to the present day, recognizing its continuity.
CO5	examine the role of language and linguistics in preserving and transmitting knowledge and cultural heritage.
CO6	discuss the interconnectedness of philosophy, science, and spirituality in IKS, highlighting their contributions to holistic understanding.

Evaluation:

Sl. No.	Marks	Details
Formative	30	In the form of Internal Exam, Assignments, Presentations, Quizzes, Projects, Activities etc.
Summative	70	In the form of University Practical Exam

Recommended Study Material:

❖ Text book:

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❖ Reference book:

1. "Indian Knowledge Systems" by Kapil Kapoor
2. "Indian Knowledge Tradition" by K. Ramasubramanian
3. "The Advaita Worldview: God, World, and Humanity" by Anantanand Rambachan
4. "The Vedas: An Introduction to the Sacred Texts of India" by Roshen Dalal
5. "The Rigveda: 3-Volume Set" translated by Stephanie W. Jamison and Joel P. Brereton
6. "The Yajur Veda: A Rendering of the Jewel of the Vedic Hymns" translated by Devi Chand
7. "A History of Indian Philosophy" by Surendranath Dasgupta (Multi-volume work)
8. "Indian Philosophy and Modern Culture" by Satyendra Nath Dasgupta
9. "The Wisdom of the Upanishads" by A. Parthasarathy
10. "The Dhammapada: The Sayings of the Buddha" translated by Eknath Easwaran
11. "Prameya Mala" by Swami Tattvavidananda Saraswati
12. "Pramana: Vaisheshika Sutra of Kanada" translated by Satish Chandra Vidyabhusana
13. "Saṃsāya in Indian Philosophy: Text and Commentaries" by Bibhuti Baruah
14. "The Art of Reasoning" by David Kelley

❖ Web material:

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B. Tech. (Mechanical Engineering) Programme

SYLLABI

University Elective Courses

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
List of University Elective Courses under
Choice Based Credit System
Second Year Undergraduate Programme
(Effective from Academic year 2017-18)

Sr. No.	Course Code & Course Name	Department / Faculty offering the Course	Semester in which the course to be offered / Remarks
1	EC281.01: Introduction to MATLAB Programming	EC / FTE	3
2	EC282.01: Prototyping Electronics with Arduino		4
3	CE281.01: Art of Programming	CE / FTE	3
4	CE282.01: Web Designing		4
5	CL281.01: Environmental Sustainability and Climate Change	CL / FTE	3
6	CL283: SDG Handprint Laboratory		3
7	CL282.01: Basics of Environmental Impact Assessment		4
7	EE283: Python For Electrical Engineering	EE/FTE	3
8	EE286: Computer Programming For Electrical Engineering		4
9	IT281.01: ICT Resources and Multimedia	IT/FTE	3
10	IT282.01: Internet Technology and Web Design		4
11	ME281.01: Engineering Drawing	ME/FTE	3
12	ME282.01: Material Science		4
13	PH233.01: Fundamentals of Packaging	RPCP/FPH	3
14	PH238.01: Cosmetics in daily life		4
15	PD260.01: Basic Laboratory Techniques	PDPIAS/FAS	3
16	PD261: Astrophysics, Space and Cosmos-1 (ASC-1)		3
17	PD262: Astrophysics, Space and Cosmos-2(ASC-2)		4
18	NR251.01: First Aid & Life Support	NURSING / FMD	3
19	NR261.01: Life Style Diseases & Management		4
20	PT191.01: Health Promotion and Fitness	ARIP / FMD	3
21	PT192.01: Occupational Health & Ergonomics		4
22	CA224: Introduction to Web Designing	CMPICA / FCA	3
23	CA225: Programming the Internet		4
24	BM231: Banking and Insurance	I ² IM / FMS	3
25	BM241: Health Care Management		4

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
EC281.01: INTRODUCTION TO MATLAB PROGRAMMING
B TECH 3rd SEMESTER (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

The course, intended for students with no programming experience, provides the foundations of programming in MATLAB. Variables, arrays, conditional statements, loops, functions, and plots are covered. At the end of the course, students should be able to use MATLAB in their own work and be prepared to deepen their MATLAB programming skills and tackle other programming languages.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Introduction to MATLAB Basics	3
2.	Basic MATLAB Functions	8
3.	Interactive computation	8
4.	Scripts and Functions in MATLAB	8
5.	Applications	3

Total Hours (Theory): 30

Total Hours (Lab): 0

Total Hours: 30

C. Detailed Syllabus:

1.	Introduction to MATLAB Basics	3 Hours	10%
1.1	MATLAB windows, On-line help		1Hr
1.2	Input- Output, File Types		1 Hr
1.3	General commands to remember		1 Hr

2.	Basic MATLAB Functions	8 Hours	25%
2.1	Working with Arrays of Number		1Hr
2.2	Creating and Printing Simple Plots		1Hr
2.3	Creating, Saving, and Executing a Script File		2Hrs
2.4	Creating and Executing a Function File		2Hrs
2.5	Working with Files and Directories		2Hrs
3.	Interactive computation	8 Hours	25%
3.1	Matrix and Vectors		1Hr
3.2	Matrix and Array Operations		3Hrs
3.3	Creating and Using Inline Functions		2Hrs
3.4	Using Built in Functions		2Hrs
4.	Scripts and Functions in MATLAB	8 Hours	25%
4.1	Scripts Files		2Hrs
4.2	Function Files		3Hrs
4.3	Language-Specific Features		3Hrs
5.	Applications	3 Hours	15%
5.1	Solving a Linear system		1Hr
5.2	Finding eigenvalues and eigenvectors		1 Hr
5.3	Matrix Factorizations		1 Hr

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes:

Students will acquire the following skills:

- Be fluent in the use of procedural statements--assignments, conditional statements, loops, function calls--and arrays.
- Be able to design, code, and test small MATLAB programs that meet requirements expressed in English. This includes a basic understanding of top-down design.

F. Recommended Study Materials

Reference Book & Text Book:

1. Getting Started with MATLAB, Rudrapratap (IISc, Bangalore), Oxford University Press.
2. A Guide to MATLAB, Brian R Hunt, Ronald L Lipsman, Cambridge University Press.

Web Materials / Reading Materials:

1. Lecture notes
2. Handouts
3. Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
EC282.01: PROTOTYPING ELECTRONICS WITH ARDUINO
B TECH 4th SEMESTER (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

- To aware students regarding Arduino Architecture
- To aware students regarding peripherals, sensors, wireless devices interfacing & Programming

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Getting Started with Arduino	02
2.	Arduino C	06
3.	GPIO Programming	08
4.	Sensors & Actuators	06
5.	Wireless Devices	08

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	Getting Started with Arduino	02 Hours	10%
1.1	Introduction to Arduino		1 Hr
1.2	Arduino Variants		
1.3	The Arduino UNO Board		1 Hr
1.4	Electronics Components		
2.	Arduino	06 Hours	20%
2.1	C Data types		2 Hrs

2.2	Control Statements		
2.3	Functions & Arrays		
2.4	Using Modifying & Creating Arduino Libraries		2 Hrs
2.5	Advanced Coding & Memory Handling in Arduino		2 Hrs
3.	GPIO Programming	08 Hours	25%
3.1	LED Interfacing with Arduino		1 Hr
3.2	Driving 7 Segment Display using Arduino		1 Hr
3.3	Serial Communications & Soft serial Utility		2 Hrs
3.4	LCD Interfacing using Arduino		1 Hr
3.5	Matrix Keypad Interfacing using Arduino		1 Hr
3.6	Controlling Speed of DC Motor using PWM Techniques		1 Hr
3.7	Driving Stepper Motor using Arduino		1 Hr
4.	Sensors & Actuators	06 Hours	20%
4.1	Introduction to Sensors & Actuators		2 Hrs
4.2	Digital On/Off Sensors		
4.3	Analog Sensors		2 Hrs
4.4	Pulse Sensors		2 Hrs
5.	Wireless Devices	08 Hours	25%
5.1	Introduction to Wireless Devices		2 Hrs
5.2	Interfacing GSM & GPS Module with Arduino		
5.3	Interfacing Bluetooth Module with Arduino		2 Hrs
5.4	Interfacing Zigbee Module with Arduino		2 Hrs
5.5	Interfacing IR Module with Arduino		2 Hrs

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous	Quiz; Assignments, Seminars, Projects,

		Evaluation	Presentation, etc.
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- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes

- Upon completion of this course, students will understand the Arduino Architecture.
- Students can understand the programming GPIOs, Sensors, Actuators & Wireless Devices.

F. Recommended Study Materials

Reference Book & Text Book:

3. Beginning of Android ADK with Arduino by Mario
4. Arduino CookBook, Michael Margolis, O'Reilly

Web Materials / Reading Materials:

1. Lecture notes
2. Handouts
3. Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
U & P U. PATEL DEPARTMENT OF COMPUTER ENGINEERING
CE281.01: ART OF PROGRAMMING
B TECH 3RD SEMESTER (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

- To be able to understand the various data structures available in programming language and apply them in solving computational problems.
- To be able to do testing and debugging of code written programming language.
- To create students' interest for programming related subjects and to make them aware of how to communicate with computers by writing a program.
- To foster the ability of solving various analytical and mathematical problems with algorithms within students.
- To make them learn regarding different data structures and memory management in the programming language.
- To promote skills like Development of logic and implementation of basic mathematical and other problems at individual level.
- To make them learn and understand coding standards, norms, variable naming conventions, commenting adequately and how to form layout of efficient program.
- To explain them concepts of pointer & file management concepts.

B. Outline of the Course:

Sr. No	Title of the Unit	Minimum number of Hours
1.	Introduction to Computer Systems	02
2.	Data Storage and Operations	03
3.	Algorithms and Flow charting	04
4	Algorithm to Program	06
5	Loops and Controls Construct	04
6	Errors and Debugging	04
7	Structured Programming	04
8	Coding Conventions	03

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	Introduction to Computer Systems	02 Hours	7%
1.1	Basic computer organisation, operating system, editor, compiler, interpreter, loader, linker, program development.		
2.	Data Storage and Operations	03 Hours	10%
	Various data representation techniques, data types, constants, variables (local and global), arrays, various arithmetic and logical operations in a typical programming environment, working with numbers, String and operators, finding pattern in string.		
3.	Algorithms and Flow charting	04 Hours	13%
	Introduction to computer problem solving, concepts and algorithms and flow chart, tracing of an algorithms, writing conditional code.		
4	Algorithm to Program	06 Hours	21%
	Specifications, top down development and stepwise refinement as per programming environment needs. Imperative style of correct and efficient programming, introductory concepts of time and space complexities.		
5	Loops and Controls Construct	04 Hours	13%
	Conditional and unconditional execution. Simple versus nested controls. Various aspects of repetitive executions, iterative versus recursive programming styles, assertions and loop invariants.		
6	Errors and Debugging	04 Hours	13%
	Types of errors, error handling, debugging, tracing/stepwise execution of program, watching variables values in memory.		
7	Programming	04 Hours	13%
	Introduction to modular approach of problem solving, concepts of procedure and functions for effective programming, Making code modular.		
8	Coding Conventions	03 Hours	10%
	Variable naming, function naming, indentation, usage and significance of comments for readability and program maintainability.		

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

F. Student Learning Outcomes

- Students will get acquainted with basic components and capabilities of a typical computing system.
- Students will be able to critically think about basic problems and develop algorithms to solve, validate and verify with computing systems.
- Students will be able to identify appropriate language constructs and approach to computational problems.
- Students will be acquainted with coding standards including documentation which are required to be used for the development of effective, efficient and maintainable programs.

F. Recommended Study Materials

Reference Book & Text Book:

- Joyce Farrell, Programming Logic and Design Comprehensive, Cenage Learning
- Sedgewick R., Algorithms in C, Addison Wesley
- V. Rajaraman, Fundamentals of Computers, Prentice Hall of India

Web Materials / Reading Materials:

- Lecture notes
- Handouts
- Chapter wise Assignment

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
U & P U. PATEL DEPARTMENT OF COMPUTER ENGINEERING
CE282.01: WEB DESIGNING
B TECH 4th SEMESTER (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

- To Learn how to link pages so that they create a Web site.
- To Design and develop a Web site using text, images, links, lists, and tables for navigation and layout.
- To Style web page using CSS, internal style sheets, and external style sheets.
- To Learn how to use graphics in Web design.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Web Designing Principles	02
2	Basic of Web Designing	03
3	Hyper Text Markup Language	06
4	Cascading Style Sheet Language	06
5	Extensible Mark-up Language	04
6	Adobe Dreamweaver	03
7	Adobe Photoshop	06

Total Hours (Theory):30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1.	Web Designing Principles	02 Hours	7%
	Basic principles involved in developing a web site, Planning process, Five Golden rules of web designing, Designing navigation bar, Page Design, Home Page Layout, Design Concept		
2.	Basic of Web Designing	03 Hours	10%
	Brief History of Internet, What is World Wide Web, Why create a web site, Web Standards and Protocols		
3.	Hyper Text Markup Language	06 Hours	20%
	Basic structure of an HTML document, Mark up Tags, Heading-Paragraphs, Line Breaks, HTML Tags and Attributes , Creating an HTML layout		
4	Cascading Style Sheets	06 Hours	20%
	Concept of CSS, Creating Style Sheet, CSS Properties, CSS Styling(Background, Text Format, Controlling Fonts), CSS Id and Class, CSS Selectors, Box Model(Introduction, Border properties, Padding, Properties, Margin properties)		
5	Extensible Mark-up Language	04 Hours	13%
	Basics of XML, XML Attribute, Elements, XML Validator		
6	Adobe Dreamweaver	03 Hours	10%
	Setting Up a Site with Dreamweaver, FTP - Site Upload Feature of Dreamweaver, Create various types of Links, Insert multimedia including text, image, animation & video		
7	ADOBE Photoshop	06 Hours	20%
	Use of selection tools, Drawing and Painting tool, Transform B/W image to color, Layers, moving layer content using move tool Manipulating Images		

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for	Quiz; Assignments,

		Continuous Evaluation	Seminars, Projects, Presentation, etc.
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- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

G. Student Learning Outcomes

Upon successful completion of this course, the student will:

- Create an Information Architecture document for a web site.
- Select and apply mark-up languages for processing, identifying, and presenting of information in web pages.
- Combine multiple web technologies to create web components.
- Incorporate best practices in navigation, usability and written content to design websites that give users easy access to the information they seek.

F. Recommended Study Materials

Reference Book & Text Book:

1. Html And CSS: Design And Build Websites, By Jon Duckett.
2. Learning Web Design: A Beginner's Guide To Html, Css, Javascript, And Web Graphics, By Jennifer Niederst Robbins
3. Beaird, Jason. 2010. The Principles Of Beautiful Web Design. 2nd Edition. Isbn-10: 098057689x
4. Adobe Dreamweaver Cs6: Classroom In A Book

Web Materials / Reading Materials:

1. Lecture notes
2. URLs such as

<https://www.w3.org/>

<http://www.land-of-web.com/category/tutorials>

<http://www.bypeople.com/design-trends-for-2011/>

<https://www.w3schools.com/>

<https://www.html5rocks.com/en/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI S. PATEL DEPARTMENT OF CIVIL ENGINEERING
CL28L01: ENVIRONMENTAL SUSTAINABILITY AND CLIMATE CHANGE
3RD SEMESTER (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objectives of the Course:

The main objectives of the course are:

- To provide a basic understanding of the major environmental problems that need to be addressed to ensure sustainable development
- To provide a basic understanding about various management approaches towards a sustainable development
- To introduce students to the environmental aspects of specific industrial sectors, such as energy, transport, land and water use, and the built environment
- To provide basic understanding about climate changes, their causative factors and the possible mitigation

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introducing Sustainability Basics and Environmental Management	03
2	Environmental Challenges	03
3	Principles of Environmental Management	08
4	Environmental Sustainability	04
5	Introduction to Climate Change	07
6	Climate Change-Mitigation	05

Total Hours (Theory): 30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1	Introducing Sustainability Basics and Environmental Management	03 Hours	10%
1.1	What is Unsustainable?		
1.2	What is Sustainability? Defining the Terms		
1.3	Development & Environment		
1.4	Environmental Strategy: The New Business Playing Field		
1.5	Environmental Management		
2	Environmental Challenges	03 Hours	10%
2.1	Depletion of Water Resources		
2.2	Population		
2.3	Agriculture		
2.4	Land Degradation		
2.5	Energy Security		
3	Principles of Environmental Management	08 Hours	26%
3.1	Environmental Concerns in India		
3.2	International Environmental Movement		
3.3	Definition, Goals, Need, Tools of Environmental Management		
3.4	Participants in EM		
3.5	Ethics and the Environment		
3.6	Ecology and the Environment		
3.7	Environmental Management Systems & Standards		
4	Environmental Sustainability	04 Hours	14%
4.1	Strategies for Sustainability		
4.2	Land Use and Urban Planning		
4.3	Energy and Climate Change		
4.4	Transportation		
4.5	Balancing Population with Food and Water Resources		
5	Introduction to Climate Change	07 Hours	23%
5.1	Climate Change-Way & Means		
5.2	What Do We Know and Don't Know?		
5.3	The Physical Science of Climate Change		
5.4	Causes of Climate Change		
5.5	Global Atmospheric Composition		
5.6	Greenhouse Gases and Aerosols		
5.7	Extreme Weather Events & Sea Level Rise		
5.8	Climate Projections and their Uncertainties		

6 Climate Change-Mitigation

05 Hours 17%

- 6.1 Global Carbon Cycle
- 6.2 Concept of Carbon Sequestration
- 6.3 Carbon Credits and Carbon Footprints
- 6.4 Policy Perspective: UNFCCC, IPCC, Kyoto Protocol, MoEFCC

D. Instructional Method and Pedagogy:

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Students Learning Outcomes:

On the completion of the course the students will be able to:

- Understand & appreciate for the value of quantitative, systems and trans disciplinary thinking of Environmental Sustainability
- Expand their awareness about the environment as an increasing part of the core business model and day-to-day operations of many organizations
- Develop an environmental blueprint for action
- Think strategically and act entrepreneurially to create sustainable future
- Review on Climate Change and related strategies

F. Recommended Study Materials:

Text Books:

1. Environmental Management, T. V. Ramchandra & Vijay Kulkarni, Teri Press, New Delhi, 2009.

2. Handbook of Environmental Laws, Acts, Guidelines, Compliances & Standard Policy, R. K. Trivedy, B.S. Publishers, 2010.
3. Climate Change & India, Vulnerability Assessment and Adaption, P. R. Shukla, University Press, Hyderabad, 2003.

Reference Books:

1. Environmental Management, Principles and Practice, C. J. Barrow, Psychology Press, 1999.
2. Environmental Management in Practice, Nath B., Hens, L., Compton, P. and Devuyst, D, Vol I, Routledge, London and New York, 1998.
3. Handbook of Environmental Management and Technology: Gwendolyn Holmes, Ben Ramnarine Singh, and Louis Theodore, Wiley, 2004.
4. Corporate Environmental Management: Welford R, University Press, Hyderabad, 1999.

Web Materials:

1. <http://nptel.ac.in/courses/122102006/7>
2. <http://envfor.nic.in/>
3. <http://cpcb.nic.in/>
4. <http://gpcb.gov.in/>
5. <http://nptel.ac.in/courses/119106008/40>
6. <https://unccelearn.org/course/>
7. <http://www.open.edu/openlearn/nature-environment/the-environment/climate-change/content-section-0>
8. <http://www.openlearningworld.com/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
MANUBHAI S. PATEL DEPARTMENT OF CIVIL ENGINEERING
CL282.01: BASICS OF ENVIRONMENTAL IMPACT ASSESSMENT
4TH SEMESTER (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objectives of the Course:

The main objectives of the course are:

- To provide a basic understanding of the need, objective and the parameters considered for Environmental Impact Assessment (EIA) studies
- To provide an awareness on impact on resources and environment from development projects
- To introduce students to the legal, economic, administrative and technical process of preparing and/or evaluating environmental impact documents
- To learn laws related to EIA and auditing in India

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1	Introduction to EIA	05
2	Considerations for Environmental Assessment	07
3	Process of Impact Assessment	07
4	Tools for Assessing Environmental Impact	07
5	EIA in Indian Scenario	04

Total Hours (Theory): 30

Total Hours (Lab): 00

Total Hours: 30

C. Detailed Syllabus:

1 Introduction to Environmental Impact Assessment	05 Hours	17%
1.1 What is EIA?		
1.2 Objectives of EIA		
1.3 Brief History of Environmental Impact Analysis,		
1.4 EIA as research,		
1.5 EIA as decision making process,		
1.6 EIA in Global Affairs		
 2 Considerations for Environmental Assessment	 07 Hours	 23%
2.1 Assessment Methodology		
2.2 Socioeconomic Impact Assessment		
2.3 Air Quality Impact Analysis		
2.4 Noise Impact Analysis		
2.5 Energy Impact Analysis		
2.6 Water Quality Impact Analysis		
2.7 Vegetation And Wild Life Impact Analysis		
2.8 Cumulative Impact Assessment		
2.9 Ecological Impact Assessment		
2.10 Risk Assessment		
 3 Process of Impact Assessment	 07 Hours	 23%
3.1 Process for Environmental Impact Study		
3.2 Terms of Reference		
3.3 Stages in EIS Production: Screening, Scoping, Prediction, Evaluation, Reducing Impact, Monitoring, Conclusions		
3.4 Components of EIA Reports		
 4 Tools for Assessing Environmental Impact	 07 Hours	 23%
4.1 Impact Assessment Methodologies - various Methods - Their Applicability		
4.2 Rapid EIA		
4.3 Strategic Impact Assessment		
4.4 Cumulative Impact Assessment		
 5 EIA in Indian Scenario	 04 Hours	 14%
5.1 Provisions in the EIA Notification by MoEFCC		
5.2 Categorization of Industries for Seeking Environmental Clearance		
5.3 Procedure for Environmental Clearance		
5.4 Environmental Management Plan		
5.5 Case Study: Sardar Sarovar Dam, Narmada Project		

D. Instructional Method and Pedagogy:

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Students Learning Outcomes:

On the completion of the course the students will be able to:

- Decide on the typical parameters to be considered in a EIA study of a developmental project
- Fully participate in interdisciplinary environmental report preparation teams
- Review and understand an EIA document for completeness and adequacy
- Review a typical development project plan and identify possible environmental effects and prepare appropriate initial studies
- Utilize EIA documents for policy development, project planning or for legal or political action planning

F. Recommended Study Materials:

Text Books:

- Environmental Impact Assessment, Larry W Canter (2nd Edition), McGraw Inc., Singapore, 1996.
- Environmental Impact Assessment – Handbook: John G Rau and D C Wooren, Mc-GrawHill.
- Environmental Impact Assessment, A. K. Shrivastava, APH Publishing, New Delhi, 2003.

Reference Books:

- Eccleston, H.C. Environmental Impact Statements. John Wiley & Sons, Inc. Canada, 2000.

2. World Bank, Environmental Assessment Sourcebook. Volume 1. World Bank Technical Paper No. 139, Washington, D.C, 1991.
3. Environmental Impact Analysis - a Decision Making Tool: By R K Jain, L. V. Urban and G.S. Stacey Publishers: Van Nostrand Reinhold New York.

Web Materials:

1. http://eia.unu.edu/course/index.html%3Fpage_id=173.html
2. <http://envfor.nic.in/legis/eia/eia-2006.htm>
3. <http://envfor.nic.in/>
4. http://nptel.ac.in/Clarify_doubts.php?subjectId=120108004&lectureId=5
5. <http://cpcb.nic.in/>
6. <http://gpcb.gov.in/>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY
IT281.01: ICT RESOURCES AND MULTIMEDIA
3RD SEMESTER (UNIVERSITY ELECTIVE)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

The main objectives for offering the course ICT RESOURCES AND MULTIMEDIA are:

1. To provide you the conceptual and technological developments in the field of information technology with the emphasis on comprehensive knowledge of Internet.
2. To introduce the fundamental elements of multimedia.
3. Understand to use the packages of presentation in detail.
4. Understand the impact of Information Technology on the Society and Various applications.

B. Outline of the Course

Sr. No.	Title of the Unit	Minimum Number of Hours
1	ICT Utilities and Tools	8
2	Introduction to Multimedia	4
3	Presentation Package	6
4	Networking Concepts	6
5	Information Technology and Society	6

Total hours (Theory): 30

Total hours (Lab): 00

Total hours: 30

C. Detailed Syllabus:

Following contents will be delivered to the students during laboratory sessions.

- | | | | |
|----|--|-----------------|-------------|
| 1. | ICT Utilities and Tools | 08 Hours | 30% |
| | <p>Compression Utilities: WinZip, PKZIP, Concept of compression, Defragmenting Hard, disk using defrag, Scan Disk for checking disk space, lost files and recovery, Formatting Hard disk, Setting System Date and Time, Antivirus.</p> <p>Tools: Prezi, Macromedia Director & Flash, Gliffy, Edmodo, Google Classroom, Glogster.</p> | | |
| 2. | Introduction to Multimedia | 04 Hours | 10 % |
| | <p>What is multimedia, Components of multimedia, Web and Internet multimedia applications, Transition from conventional media to digital media.</p> | | |
| 3. | Presentation Package | 06 Hours | 20% |
| | <p>Microsoft PowerPoint: Slide layout, Slide design (Proper selection based on audience), Header and Footer in slides, Slide transition, Slide Master, Insert Picture-Smart Art, Insert animations to different objects, Hide Slide, Rehearse Timings, Record slide show. How to prepare professional presentation.</p> | | |
| 4. | Networking Concepts | 06 Hours | 20% |
| | <p>What is Networking, Local Area Networking (LANs), Metropolitan Area Network , MAN), Wide Area Network (WAN), Networking Topologies, Transmission media & method of communication.</p> | | |
| 5. | Information Technology and Society | 06 Hours | 20% |
| | <p>Indian IT Act, Intellectual Property Rights – issues. Application of information Technology in Railways, Airlines, Banking, Insurance, Inventory Control, Financial systems, Hotel management, Education, Video games, Telephone exchanges, Mobile phones, Information kiosks, special effects in Movies.</p> | | |

D. Instructional Method and Pedagogy:

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes:

At the end of the course the students will be able to:

- A student will be having the basic knowledge of the Information Technology.
- A student will be able to understand effectively use miscellaneous utilities such as: Compression, CD writing, Antivirus etc.
- A student will be able to use different tools for analysis and presentation.
- A student will be able to know the impact of ICT tools and Information Technology in Society.

F. Recommended Study Material:

Text Books:

- ITL Educational Society, "Introduction to IT", Pearson Education, 2009.
- William Stallings, "Data and Computer Communication", Prentice, Hall of India Private Limited.
- Mavis Beacon, "All-in-one MS Office" CD based views for self-learning, BPB Publication, 2008
- Li & Drew, "Fundamentals of Multimedia", Pearson Education, 2009.

Reference Books:

- Tay Vaughan, "Multimedia making it work", Tata McGraw-Hill, 2008.
- Parekh Ranjan, "Principles of Multimedia", Tata McGraw-Hill, 2007.

Web Materials:

- <https://www.tutorialspoint.com>
- <http://computingcareers.acm.org>

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF INFORMATION TECHNOLOGY
IT282.01: INTERNET TECHNOLOGY AND WEB DESIGN
4TH SEMESTER (UNIVERSITY ELECTIVE)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	02		02	2
Marks	00	100	100	

A. Objective of the Course:

The main objectives for offering the course software project are:

- To describe the basic concepts for web development.
- To provide exposure in the field of Internet Technologies and Web Development.
- Learn the basic working scheme of the Internet and World Wide Web.
- To provide an intuition in web designing and development using Hyper Text Markup Language (HTML).
- Specify design rules in constructing web pages and sites.

B. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to WWW	03
2.	Current Trends on Internet	04
3.	HTML Programming Basics	06
4.	Web Publishing and Browsing	09
5.	Services on Internet	04
6.	Electronic Mail	04

Total hours (Theory): 30

Total hours (Lab): 00

Total hours: 30

C. Detailed Syllabus:

1. Introduction to WWW 03 Hours 10%
Web Related browser functions, Browser Configuration, Browser Security

Issues, Cookies, Spider, Intelligent Agents and special purpose browsers.

2. **Current Trends on Internet** **04 Hours** **13.33%**
Introduction, Languages, Internet Phone, Internet Video, collaborative computing, e-commerce.
3. **HTML Programming Basics** **06 Hours** **20%**
HTML page structure, HTML Text, HTML links, HTML document tables, HTML Frames, HTML Images, multimedia.
4. **Web Publishing and Browsing** **09 Hours** **30%**
Overview, SGML, Web hosting, Documents Interchange Standards, Components of Web Publishing, Document management, Web Page Design Consideration and Principles, Search and Meta Search Engines, Publishing Tools.
5. **Services on Internet** **04 Hours** **13.33%**
E-mail, WWW, Telnet, FTP, IRC and Search Engine.
6. **Electronic Mail** **04 Hours** **13.33%**
Email Networks and Servers, Email protocols –SMTP, POP3, IMAP4, MIME6, Structure of an Email – Email Address, Email Header, Body and Attachments, Email Clients: Netscape mail Clients, Outlook Express, Web based E-mail. Email encryption- Address Book, Signature File.

D. Instructional Method and Pedagogy:

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcome:

At the end of the course the students will be able to:

- Review the current topics in Web & Internet technologies.
- Learn the basic working scheme of the Internet and World Wide Web.
- Understand fundamental tools and technologies for web design.
- Comprehend the technologies for Hypertext Mark-up Language (HTML).
- Specify design rules in constructing web pages and sites.

F. Recommended Study Material:

Text Books:

1. HTML 4 , Bryan Pfaffenberge, Bible.
2. Greenlaw R and Hepp E “Fundamentals of Internet and www” 2nd EL, Tata McGrawHill,2007.
3. Comer, “The Internet Book”, Pearson Education, 2009

Reference Books:

1. M. L. Young,”The Complete reference to Internet”, Tata McGraw Hill, 2007.
2. Godbole AS & Kahate A, “Web Technologies”, Tata McGrawHill,2008.
3. Jackson, “Web Technologies”, Pearson Education, 2008.
4. Leon and Leon, “Internet for Everyone”, Vikas Publishing House.

Web Materials:

1. www.w3schools.com
2. www.tutorialspoint.com

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
ME 281.01 ENGINEERING DRAWING
B TECH 3RD SEMESTER (UNIVERSITY ELECTIVE)

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

- To introduce the student to the universal language and tool of communication of engineers.
- To make them thorough in understanding and using the various concepts of Engineering Drawing.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Engineering Drawing	05
2.	Visualization of Objects	10
3.	Sectional View of Objects	08
4.	Assembly and Detailed Drawing	04
5.	Introduction to Computer Aided Drawing	03

Total hours (Theory): 30

Total hours (Lab): 00

Total: 30

C. Detailed Syllabus:

- | | | |
|--|----------|-----|
| 1 Introduction to Engineering Drawing | 05 Hours | 17% |
| 1.1 Basic of engineering graphics. | | |
| 1.2 Lines, types of lines, use of different line in engineering drawing. | | |
| 1.3 Dimensioning, placing of dimensions, general rules of dimensions. | | |

- 2 **Visualization of Objects** 10 Hours 34%
- 2.1 Principle of visualization of objects, need of visualization, methods of visualization of three dimensional objects.
- 2.2 Interpretation of line and area in orthographic drawing.
- 2.3 Visualization of objects -pictorial view and orthographic view.
- 2.4 Visualization of objects- Isometric view.
-
- 3 **Sectional view of Objects** 08 Hours 27%
- 3.1 Principle of sectional view of objects,
- 3.2 Classification of sectional views,
- 3.3 Full sectional views, half sectional views, convention for sectioning.
-
- 4 **Assembly and Detailed Drawing** 04 Hours 12%
- 4.1 Assembly drawing with sectioning from given detailed
- 4.2 Detail drawing from given assembly
-
5. **Introduction to Computer Aided Drawing** 03 Hours 10%
- 5.1 Introduction to 2D Drawing facilities in CAD software
- 5.2 Introduction to 3D modeling & its relationship with 2D drawing views

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

E. Student Learning Outcomes / objectives:

Upon successful completion of this course, the student will be able to:

1. Understand the basics of drawing which is used in industries.
2. To convert sketches to engineered drawings will increase.
3. Improve their visualization skills so that they can apply these skills in developing new products.
4. Know the fundamental of Computer Aided Drawing & 3D Modeling.

F. Recommended Study Material:

Text Books:

1. Shah, P. J., Engineering Drawing Vol. I & II, S. Chand & Co.
2. Bhatt, N. D., Engineering Drawing, Charotar Publishing House

Reference books:

8. Gopal Krishna K.L., Engineering Drawing, Subhas Publications
9. Venugopal, K., Engineering Drawing made Easy, Wiley Eastern Ltd.
10. Agrawal, M.L. & Garg, R.K., Engineering Drawing Vol-I, Dhanpatrai & Co.
11. French, T.E., Vierck, C.J. & Foster, R. J., Graphic Science and Design, McGraw Hill
12. Luzadder, W. J. & Duff, J.M., Fundamentals of Engg. Drawing, Prentice Hall
13. Venugopal, K., Engg. Drawing and Graphics, New Age international Pry. Ltd.

Web Materials:

1. users.rowan.edu/~eyerett/courses/lfrclil/Lectures/Draw.ppt
2. mechanical-engineering-drawing.ppt.fyxm.net

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF MECHANICAL ENGINEERING
ME282.01: MATERIAL SCIENCE
B TECH 4TH SEMESTER (UNIVERSITY ELECTIVE)

Credits Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective of the Course:

- To give an overview on engineering requirements of materials and the importance of structure property relations of engineering materials on their selections and use.
- To understand the testing/ evaluation of strength and other properties of engineering materials.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum number of hours
1	Introduction to Material Science	05
2.	Metals and Alloys	09
3.	Non-Metallic Materials	08
4.	Materials Testing and Inspection	08

Total hours (Theory): 30

Total hours (Lab): 00

Total: 30

C. Detailed Syllabus:

- | | | |
|--|----------|-----|
| 1 Introduction to Material Science | 05 Hours | 16% |
| 1.1 Classification of engineering materials | | |
| 1.2 Engineering requirements of materials | | |
| 1.3 Properties of engineering materials | | |
| 1.4 Criteria for selection of materials for engineering applications | | |
| 1.5 Concept of electronic materials, classification, properties and application of electronics materials | | |

2 Metals & Alloys 09 Hours 34%

2.1 Classification of Metals

2.2 Ferrous Metals: Classification, Composition, Properties, Application (for plain carbon steel, alloy steel including stainless steel and cast iron)

2.3 Non-Ferrous Metals: Classification, Composition, Properties, Application (for copper, copper alloys, aluminum and aluminum alloys)

3 Non-Metallic Materials 08 Hours 25%

3.1 Introduction and classification of non-metallic materials

3.2 Classification of polymers on basis of thermal behavior (thermoplastics & thermosetting), properties and applications of polymers

3.3 Composites: Introduction, Characteristics of composites, Types and applications of composites

3.4 Other non-metallic materials-types, properties and applications. (like plastic, rubber, ceramics, refractories, insulators, wood etc.)

4. Materials testing and inspection 08 Hours 25%

4.1 Introduction to Non-destructive testing

4.2 Radiography Testing

4.3 Dye Penetration Testing, Magnetic Particle Testing

4.4 Ultrasonic Testing

4.5 Macro-examination, Spark Test, Macro-etching, Fracture, Microscopic examinations

D. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

E. Student Learning Outcomes:

Upon successful completion of this course, the student will be able to:

1. Understand basic engineering properties of materials and its Application.
2. Understand fundamentals of various engineering steels and alloys and should have ability to select suitable materials based on their properties.
3. Classify non-metallic materials.
4. Examine various engineering materials.

F Recommended Study Material:

Text Books

1. Narang G. B. S. and Manchanedy K., “Materials and Metallurgy”, Khanna Pub New Delhi, India
2. Dharmendra K umar and Jain S. K., “Material science and manufacturing process”, Vikas Pub House, New Delhi, India
3. R.S.Khurmi and R.S.Sedha, “Material Science”, S.Chand
4. D.S.Nutt, “Materials science and metallurgy”, S.K.Katariya and sons, Delhi

Reference Book:

1. Raghvan V., “Metallurgy for engineers”, Prentice Hall of India
2. Khanna O. P., “Material Science - A Text Book of Material Science & Metallurgy”, Dhanpat Rai Pub

Reading Materials, web materials with full citations:

1. <http://ocw.mit.edu/OcwWeb/web/courses/courses/index.htm#MaterialsScienceandEngineering>
2. http://nptel.iitm.ac.in/courses/Webcourse-contents/IIScBANG/Material%20Science/New_index1.html

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF PHARMACY
University Level Elective for Undergraduate Students of CHARUSAT
SEMESTER III
Fundamentals of Packaging [PH233.01]

Credit and Schemes:

Credits	Teaching Scheme	Evaluation Scheme				
	Contact Hr/Week	Theory		Practical		Total
		Internal	External	Internal	External	
02	02	00	00	30	70	100

A. Course Objectives

The course is structured to introduce the students, with diversified background, to different types of packaging materials generally employed and evaluation methods to be adopted for those packages.

B. Methodology and Pedagogy:

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

C. Learning Outcome:

Up on completion of the course, students would be able to,

- Describe and identify different types of packages
- Be able to describe significance of tests to be performed to evaluate packages as well to suggest the type of tests to be performed for different type of packages
- Suggest the type of material likely to be adopted for particular product
- Describe the steps involved in aseptic manufacturing and packing of products

5. Able to identify different component of aerosol packages and be able to describe role of each of those component

D. Outline of the Course

Sr No.	Unit
1	Introduction to Packaging
2	Packaging of oral solid formulation
3	Sterilization and Sterile Products Packaging including Aerosol Packaging

E. Detailed Syllabus

Sr.No	Units
1	Introduction to Pharmaceutical Packaging Definition, introduction to packaging, role of packaging, components of packaging, Overview of the Packaging Development and various aspects of it, Evaluation of packages and Physicochemical characteristics.
2	Packaging of Oral Solids Flexible Packaging Materials – Introduction to Plastic, Cellulose and Semi Synthetic Polymeric Materials, Overview of Packaging of Tablets, Capsules and Powders.
3	Sterilization and Sterile Products Packaging Over view of Sterilization Processes, Aseptic Packaging, Packaging Systems, Assessment of Sterility. Aerosol Packaging

F. Recommended Study Material

Core Books

1. Encyclopedia of Pharmaceutical Technology Vol.1-3, Swarbric, J and Bolyln, J. C., Marcel Dekker, Inc., New York.
2. Smart Packaging Technologies for fast moving consumer goods, Editor Joseph Kerry and Paul Butler, Wiley
3. Pharmaceutical Packaging Technology, Dean, D. A. Evans, E. R. and Hall, j. H., Taylor and Francis, London.
4. Handbook of Packaging Technology, by Eiri Board (Engineers India Research Institute).

Reference Books

1. Packaging of Pharmaceutical & Healthcare products, H. Lockhart, F. A. Paine, Champman and Hall, London.
2. Packaging of Pharmaceuticals, C.F. Ross, Newnes-Butterworth.
3. The Theory and Practice of Industrial pharmacy, Lachmann, L., Lieberman, H.A. & Kanig, J.I., Lea and Fibiger, CBS Publishers and Distributors, New Delhi.

Web References

1. <http://www.creativebloq.com/branding/packaging-design-resources-4131480>
2. <https://www.greenerpackage.com/newsletter>

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF PHARMACY
University Level Elective for Undergraduate Students of CHARUSAT
SEMESTER IV
Cosmetics in Daily Life [PH238.01]

Credit and Schemes:

Credits	Teaching Scheme	Evaluation Scheme				
	Contact Hr/Week	Theory		Practical		Total
		Internal	External	Internal	External	
02	02	00	00	30	70	100

A. Course Objectives

The course is structured to introduce the students, with diversified background, for preparation and evaluation of commonly used cosmetic formulations. The course will also impart preliminary information about the mechanism through which these products would act.

B. Methodology and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

C. Learning Outcome

Up on completion of the course, students would be able to,

- Describe the current market scenario of cosmetics
- Describe the various type of cosmetics for skin and hair care
- Suggest the type of dosage forms likely to be adopted for particular product

4. Describe the evaluation, packaging and labeling of various skin and hair care products.

D. Outline of the Course

Sr. No.	Unit
1	Introduction to Cosmetics and Market Scenario
2	Skin care products
3	Hair care products

E. Detailed Syllabus

Sr. No	Units
1	Introduction to Cosmetics and Market Scenario Definition, Classification, Market Scenario in India and World Wide, Introduction to Solution, Suspension, Emulsion etc.
2	Skin care products Anatomy and physiology of skin, classification of various skin care products. Formulation, evaluation, packaging and labeling of various skin care products like skin creams and lotions, suntan and anti-sunburn, skin bleaching, skin tonics, anti-aging cream.
3	Hair care products Anatomy and physiology of hair, classification of various hair care products. Formulation, evaluation, packaging and labeling of various hair care products like shampoo, conditioner, hair tonics, hair wave sets.

F. Recommended Study Material

Core Books

1. Butler, H. ed., 2013. Poucher's perfumes, cosmetics and soaps. Springer Science & Business Media.
2. Cosmetics Formulation Manufacturing & Quality Control, P.P.Sharma, 4th Ed., Vandana Publications.
3. Handbook of Cosmetic Science and Technology, Andre Barel, Marc Paye, Howard I. Maibach, CRC Press.
4. Cosmetic technology, Nanda S, Nanda A, Khar RK., Birla Publications Pvt. Ltd.
5. Cosmetics: Science and Technology, Balsam S.M. and Sagarin Edward, 2nd Ed, Wiley Interscience.

Reference Books

1. Encyclopedia of Pharmaceutical Technology Vol.1-3, Swarbric, J and Bolyln, J. C., Marcel Dekker, Inc., New York.
2. Draelos, Z.D. ed., 2015. Cosmetic dermatology: products and procedures. John Wiley & Sons.
3. The Theory and Practice of Industrial pharmacy, Lachmann, L., Lieberman, H.A. & Kanig, J.I., Lea and Fibiger, CBS Publishers and Distributors, New Delhi.

Web References

1. www.cir-safety.org/
2. <https://www.fda.gov/Cosmetics/>
3. <https://www.cosmeticseurope.eu/cosmetics>
4. www.cdsc0.nic.in/

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
University Elective (UG)
PD260.01 BASIC LABORATORY TECHNIQUES

Credit: 02

A. Objective of the course

The objectives of this course is to introduce students to the use of various electrical/electronic instruments, their construction, applications, principles of operation, standards and units of measurements; and provide students with opportunities to develop basic skills in the design of electronic equipment.

B. Outline of the Course

- 1. Error and uncertainty in measurements** (10 Lectures)
Accuracy and precision, Significant figures, Error and uncertainty analysis, Types of errors: Gross error, systematic error, random error. Statistical analysis of data, (Arithmetic mean, deviation from mean, average deviation, standard deviation, chi-square), and curve fitting, Guassian distribution.
- 2. Basic of Measurement** (5 Lectures)
Instruments accuracy, precision, sensitivity, resolution range etc. Errors in measurements and loading effects. Multimeter: Principles of measurement of dc voltage and dc current, ac voltage, ac current and resistance. Specifications of a multimeter and their significance.
- 3. DC and AC indicating Instruments** (5 Lectures)
Ideal Constant-voltage and Constant-current Sources, Voltmeter, Ammeters, Cathode Ray Oscilloscope
- 4. Signal Generators and Analysis Instruments** (5Lectures)
Block diagram, explanation and specifications of low frequency signal generators. Pulse generator, and function generator. Brief idea for testing, specifications. Distortion factor meter, wave analysis.
- 5. Digital Instruments** (5Lectures)
Principle and working of digital meters. Comparison of analog& digital instruments. Characteristics of a digital meter. Working principles of digital voltmeter, working of a digital multimeter.

C. Instructional Methods and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks	Evaluation Methodology
External	70 Marks	Practical Examination, Presentation with Report,

		Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance
		25 Marks for Continuous Evaluation
		Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

D. Student Learning Outcomes

Upon successful completion of this course, the student will be able to (Knowledge based) identify electronics/ electrical instruments, their use, peculiar errors associated with the instruments and how to minimise such errors.

E. Recommended Study Material

1. Gupta B. R. (2003). Electronics and Instrumentation. Published by S. Chand and Company Ltd, New Delhi, India.
2. Measurement, Instrumentation and Experiment Design in Physics and Engineering, M. Sayer and A. Mansingh, PHI Learning Pvt. Ltd.
3. Experimental Methods for Engineers, J.P. Holman, McGraw Hill
4. Introduction to Measurements and Instrumentation, A.K. Ghosh, 3rd Edition, PHI Learning Pvt. Ltd.
5. Principles of Electronic Instrumentation, D. Patranabis, PHI Learning Pvt. Ltd.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE

PD261 : Astrophysics, Space and Cosmos-1 (ASC-1)

University Elective
Semester 3 Undergraduate Programme
Semester 1 Postgraduate Programme

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

There have been frontier developments in recent months and past couple of years which involve major coming together of cosmology, physics and engineering sciences, such as the gravitational waves detected by LIGO and the fascinating images of shadows of the ultra-compact objects (e.g. black hole) at the center of our neighbouring galaxy M87. This development have spurred major interest in young students, citizens as well as scientific community as a whole and major research activities have started throughout the world & internationally in the academic centres. There is a huge interest in CHARUSAT students also. To meet up this demand & to create frontier research activity in terms of projects and proposals this course Astrophysics, Space and Cosmos has been devised. These will create important dividends in terms of frontier research emerging from CHARUSAT.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Basic Astronomical Techniques	10
2.	The Frontier of Space	06
3.	The Life and Death of Stars	08
4.	The Universe	06

Total Hours (Theory): 0 Total Hours

(Lab): 30 Total Hours: 30

C. Detailed Syllabus:

1.	Basic Astronomical Techniques	10 Hrs
	Description of Telescopes, Methods of observation, electromagnetic spectral window, resolution, sensitivity, noise, signal to noise ratio, background, aberrations, Telescopes at different wavelengths, Detectors at different wavelengths, Imaging techniques, spectroscopy, calibration, Atmospheric effects at different wavelengths, Astronomical data analysis, H-R diagram, sun and solar system, Sky Gazing.	6 (L) + 4 (P)
2.	The Frontier of Space	6 Hrs
	Outer reaches of earth's Atmosphere, Earth's Atmospheric Layers and Orbital Mechanism: Types of Orbits Launching of Satellites, Basics of Satellite subsystems, channel and link. Satellite Applications: Earth Observation, Scientific Study, Weather Forecast, Military Applications, GPS.	
3.	The Life and Death of Stars	8 Hrs
	Stars and Galaxies, The story of collapsing stars: Star formation and evolution, Interstellar Nebula, Red Giant, Planetary Nebulae, Planetary Dynamics, White Dwarfs, Red super Giant, Supernovae- Neutron star, Black hole, naked singularity. Gravitational Lensing, Accretion disks Around Compact Objects, Binary pulsar, Gravitational Waves.	
4.	The Universe	6 Hrs
	Universe in different scale: solar scale, Galactic scale, Cosmological scale, vastness of our universe, Expanding universe (Big bang, inflation), some interesting facts of our universe.	

D. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Student's needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

E. Student Learning Outcomes:

Students will get preliminary knowledge about Astrophysics, Space and Cosmology. This is a level one course, which provides basis for the second level course. After completion of this course, students can start taking small project in this or allied fields.

F. Recommended Study Materials

❖ Reference Book & Text Book:

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication

**CHAROTAR UNIVERSITY OF SCIENCE &
TECHNOLOGY FACULTY OF APPLIED SCIENCES
DEPARTMENT OF PHYSICAL SCIENCE**

PD262: Astrophysics, Space and Cosmos-2 (ASC-2)

University Elective

**Semester 4 Undergraduate Programme
Semester 2 Postgraduate Programme**

Prerequisite:

Student having exposure of 10+2 level of Physics, and studying in bachelor and master program from any institute can join this course.

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	2	2	2
Marks	-	100	100	

A. Objective of the Course:

There have been frontier developments in recent months and past couple of years which involve major coming together of cosmology, physics and engineering sciences, such as the gravitational waves detected by LIGO and the fascinating images of shadows of the ultra-compact objects (e.g. black hole) at the center of our neighbouring galaxy M87. This development have spurred major interest in young students, citizens as well as scientific community as a whole and major research activities have started throughout the world & internationally in the academic centres. There is a huge interest in CHARUSAT students also. To meet up this demand & to create frontier research activity in terms of projects and proposals this course Astrophysics, Space and Cosmos has been devised. These will create important dividends in terms of frontier research emerging from CHARUSAT.

B. Outline of the Course:

Sr. No.	Title of the Unit	Minimum Number of Hours
1.	Black holes and Shadows	10
2.	Gravitational Waves	06
3.	Sensors and Detectors for Satellite Imaging	08
4.	Cosmology	06

Total Hours (Theory): 0 Total Hours

(Lab): 30 Total Hours: 30

C. Detailed Syllabus:

1.	Black holes and Shadows	10 Hrs
	Spacetime, Schwarzschild metric, Null Geodesics in Schwarzschild metric, Photon sphere, Einstein ring, Shadow of Black hole, shadow of Naked singularity, recent observation of black hole image (M87 Galaxy), experiments using imaging technique	8 (L) + 2 (P)
2.	Gravitational Waves	6 Hrs
	Introduction to Gravitational Waves; recent development for the detection of Gravitational waves; Overview of Gravitational Wave detectors (i.e. LIGO, KAGRA, etc.) and its scientific and technologic importance; Outline of Interferometers & its applications; Construction and working of Michelson Interferometer & its application potentiality in Advance- LIGO; Demonstration experiment of Michelson Interferometer.	4 (L) + 2 (P)
3.	Sensors and Detectors for Satellite Imaging	8 Hrs
	Introduction to sensors, Signal, Parameters, Types of Sensors: CCD and CMOS, CMOS sensors: Architecture Configuration and Working, CCD sensors: Architecture Configuration and Working.	
4.	Cosmology	6 Hrs
	Galaxies: General Description, The Milky-Way, M87 Galaxy, Rotation Curves of Spiral Galaxies, Galaxy Formation, Expanding Universe, Hubble's Law, Dark matter and Dark Energy.	

D. Instructional Methods and Pedagogy

The topics will be discussed in interactive class room sessions using classical black-board teaching, ICT, hands-on-experiments and demonstration of experiments, whichever is relevant. Assignments, small projects and lab-exercise will be to the students. Students needs to submit solution/report/results of above mentioned work, which will be eventually used for the evaluation purpose. Occasionally seminar/viva presentation will also be used as one of the evaluation tools.

E. Student Learning Outcomes:

After completion of Astrophysics, Space and Cosmology level one and two courses, students will explore themselves for taking projects, which may eventually fulfill their curiosity in these and allied fields. It may open up new avenues in their career.

F. Recommended Study Materials

❖ Reference Book & Text Book:

1. The Story of Collapsing Stars: Black Holes, Naked Singularities, and the Cosmic Play of Quantum Gravity, Prof. Pankaj S. Joshi
2. An Introduction to Cosmology 3ed, J.V. Narlikar
3. An introduction to modern cosmology, Andrew R. Liddle
4. Astronomy: Astronomy for Beginners: The Magical Science of Stars, Galaxies, Planets, Black Holes, Wormholes and much, much more! (Astronomy, Astronomy Textbook, Astronomy for Beginners), Miles Clarke
5. Satellite Communication, Dennis Roddy, Mc-Graw Hill publication
6. Satellite Communication, Timothy Pratt, Charles Bostian, Jeremy Allnut, John Wiley and Sons publication.
7. CMOS/CCD Sensors and Camera Systems, Second Edition, Gerald Holst, Terrence S Lomheim, SPIE publication.
8. Review article: Advanced LIGO, The LIGO Scientific Collaboration, Classical and Quantum Gravity, Volume 32, Number 7 (2015)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ManikakaTopawala Institute of Nursing
University Level Elective for Undergraduate students:
NR 251.01- First Aid & Life Support (FALS)

Credits & Scheme:

Semester	Credits	Teaching Scheme	Evaluation Scheme				
		Contact	Theory		Practical		Total
		Hours/Week	Internal	External	Internal	External	
III	02	02	00	00	30	70	100

A. Course Objectives

Upon completing the course, students will be able to

- Demonstrate basic first aid skills needed to control bleeding and immobilize injuries.
- Demonstrate the skill needed to assess the ill or injured person.
- Demonstrate skills to assess and manage foreign body airway obstruction in infants, children and adults.
- Demonstrate skills to provide one- and two- person cardiopulmonary resuscitation to infants, children and adults

B. Course Outline

Unit No.	Title of Unit	Prescribed Hours
1.	Introduction and Basics of First Aid: • Rescuer Duties, Victim and Rescuer Safety • Looking for Help • After the emergency	2
2.	Medical emergencies and their first aid: • Breathing Problems • Choking in an Adult • Allergic Reactions • Heart Attack • Fainting • Diabetes and Low Blood Sugar • Stroke • Shock Abdominal maneuver, CPR, ventilation etc.	12

3.	Injuries emergencies and their first aid: • Bleeding: You Can See/ You Can't See • Wounds • Burns and Electrical Injuries • Fractures Bandaging, immobilization, transferring etc.	8
4.	Environmental Emergencies and first aid: • Bites and Stings • Poison Emergencies • Heat-Related Emergencies • Cold-Related Emergencies	7
5.	Preparation of First Aid Kit	1
Total		30 Hours

C. Instruction Method and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

D. Evaluation: The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sr. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class	Minimum 80%		10

	Participation	attendance	
		Total	30

External Evaluation

The University Theory examination will be of 70 marks and will test the logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Theory Paper	01	70	70
		Total		70

E. Learning Outcomes:

At the end of the course, learners will be able to:

- Demonstrate bandaging and immobilization of patient
- Demonstrate the skill for transferring injured person
- Demonstrate skills to clear airway & ventilate the patient
- Demonstrate skills to perform CPR

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ManikakaTopawala Institute of Nursing
University Level Elective for Undergraduate students:
NR 261.01- Life Style Diseases & Management (LSDM)

A. Credits & Scheme

Semester	Credits	Teaching Scheme	Evaluation Scheme				
		Contact	Theory		Practical		Total
		Hours/Week	Internal	External	Internal	External	
IV	02	02	00	00	30	70	100

B. Course Objectives

Upon completing the course, students will be able to

- Describe the National Health Programmes (NPDCS and NCCP) related to non-communicable diseases
- Understand the risk factors associated with non-communicable diseases and lifestyle disorders
- Understand the concept of health promotion
- Plan, implement, monitor and evaluate a health promotion programme
- Management of life style disorders

C. Course Outline:

Unit No.	Title of Unit	Prescribed Hours
1.	Introduction of NCDs/life-style disorders: • Overview of life-style disorders and epidemiology • Burden of life-style disorders in India and World • Risk factors for life-style disorders	4
2.	Management of Non-communicable diseases: • Obesity • Hypertension • Diabetic mellitus • Cardiovascular diseases • Chronic Kidney Diseases • Cancer • Stroke • Stress	20

- Alcohol and drug abuse
3. Preventive aspects of life-style diseases 4
 - Health Promotion strategies
 - Physical activity
 - Role of diet
 4. National Health Programmes (NPDCS and NCCP) related to non-communicable diseases 2
- Total 30 Hours

D. Instruction Method and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Evaluation

The students will be evaluated continuously in the form of internal as well as external examinations. The evaluation (Theory) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignments	1	8	8
2	Internal Test/ Model Exam	1	12	12
3	Attendance and Class Participation	Minimum 80% attendance		10
Total				30

External Evaluation

The University Theory examination will be of 70 marks and will test the logic and critical thinking skills of the students by asking them theoretical as well as application based questions. The examination will avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Theory Paper	01	70	70
Total				70

F. Learning Outcomes

At the end of the course, learners will be able to:

- Describe the National Health Programmes (NPDCS and NCCP) related to non-communicable diseases
- Understand the risk factors associated with non-communicable diseases and lifestyle disorders
- Understand the concept of health promotion
- Plan, implement, monitor and evaluate a health promotion programme
- Management of life style disorders

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
Semester III (University Elective)
PT191.01 HEALTH PROMOTION&FITNESS

Credit Hours

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
02			02			00	100	100

A. Objectives of The Course

This course will introduce students to the basic concept of health promotion, fitness and screening and basic assessment of fitness.

B. Outline of The Course

Sr. No.	Title of the unit	Minimum number of hours
1.	Basic Concept of Health Promotion	12
2.	Epidemiology and Health Promotion in Different Setting	12
3.	Basic Concept of Fitness	12
4.	Fitness Assessment	12

Total hours (Theory) : 48

Total hours (Practical): 00

Total hours: 48

C. Detailed Syllabus

- | | | |
|-----|---|--------------|
| 1 | Basic Concept of Health Promotion | 12hrs |
| 1.1 | Meaning of health and Wellness | |
| 1.2 | Cultural & Social determinants of Health | |
| 1.3 | Physical, Environmental, Emotional & Psychological health | |
| 1.4 | Promotion of Healthy Lifestyles through Physical Activity, Diet, Stress Management, | |

- Avoiding Tobacco – Alcohol
- 1.5 Promotion of Personal Hygiene, Treatment Seeking Behavior, Treatment Compliance and Reducing Stigma
- 1.6 Need of health promotion in India
- 2 Epidemiology and Health Promotion in Different Setting 12hrs**
- 2.1 Health Statistics: Analysis and Interpretation of Data Related to Health Promotion
- 2.2 Use of Health Management Information System and Information Technologies in
- 2.3 Health Promotion
- 2.4 Health promotion in different settings - emergency and disaster
Different areas of health promotion in India as compared to developed countries
- 3 Basic Concept of Fitness 12 Hrs**
- 3.1 Introduction definition of term :Fitness
- 3.2 Basic Concepts Of Fitness
- 3.3 Mental and physical fitness
- 3.4 Health benefits of activity and Fitness
- 4 Fitness Assessment 12Hrs**
- 4.1 Multifactorial fitness assessment and screening : Physical activity screening :Identify risk factors, height, weight, BMI, Physically active hours
- 4.2 Aerobic fitness, Muscular Fitness, Activity and Weight control
Vitality and Longevity
- 4.3 Clinical preventive screening for infants
- 4.4 Nutritional screening

D. Instructional Method and Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks	Evaluation Methodology
External	70 Marks	Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.

Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Student Learning Outcomes

At the end of the semester the student will be able:

1. To review the basics health, Promotion & Fitness
2. To understand the need of Health Promotion
3. To understand basic concept of fitness
4. To develop necessary skill to screen and assess basics of fitness

F. Recommended Study Material

Text Books

1. Textbook of Preventive & Social Medicine- Dr. K. Park
2. Textbook of community medicine: V. K. Mahajan
3. Chiropractic, Health, Promotion and Wellness – Meridell Gatterman MA, DC, Med
4. Health, Promotion and Wellness :evidence based guide to clinical preventive services—Cheryl Hawk & Will Evas
5. Fitness and Health – 6th edition – Brian J Sharkey, PhD

Reference Books

1. Principles Of Health Education And Health Promotion, (2nd edition), J. Thomas Butler, Morton Publishing Company, Englewood, Colorado
2. Foundations Of Health Education, R. M. Eberst, Editor, Coyote Press, San Bernardino: 1998-99
3. Evaluation in health promotion – principles and perspective- WHO Regional Publications, European Series, No. 92
4. Principles and foundation of health promotion and education (5th edition) by Randall R. Cottrell, James T. Girvan, James F. McKenzie

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MEDICAL SCIENCES
ASHOK & RITA PATEL INSTITUTE OF PHYSIOTHERAPY
Semester IV (University Elective)
PTI92.01 OCCUPATIONAL HEALTH & ERGONOMICS

Credit Hours:

Hrs. / Wk.			Credits			Total Marks		Total
L	P	T	L	P	T	Theory	Practical	
02			02			00	100	100

A. Objectives of The Course

This course will introduce students to the basic concept of occupational health & Ergonomics Assessment and management for prevent the Occupational Hazards.

B. Outline of The Course

Sr. No.	Title of the unit	Minimum number of hours
1.	Occupational Health	06
2.	Ergonomics	12
3.	Environmental Factors & Ergonomics	06
4.	Occupational Ergonomics	12
5.	Occupational Ergonomics In Special Areas	12

Total hours (Theory) : 48

Total hours (Practical): 00

Total hours: 48

C. Detailed Syllabus

1	Occupational Health	06hrs
1.1	Introduction.	
1.2	Objective & basic concepts of Occupational health.	
1.3	Reorganization of health hazards.	

- 1.4 Early detection of Occupational diseases.
- 2 Ergonomics 12hrs**
- 2.1 Definitions of terms: Ergonomics, Ergonomists. Social significance of ergonomics.
- 2.2 Biomechanical, Physiological & Anthropometric factors related to ergonomics.
- 2.3 Ergonomical aspects of Postures like Sitting, Standing, Hand & arm postures.
- 2.4 Ergonomical aspects of movements like Pushing, Pulling, Lifting & carrying.
- 3 Environmental Factors & Ergonomics 06 hrs**
- 3.1 Noise, Light, Vibration, Climate & Chemical substances.
- 4 Occupational Ergonomics 12 hrs**
- 4.1 Work environment and control measures in work place.
- 4.2 Prevention of occupational diseases and accidents.
- 4.3 Effects of lifestyle and behavior on health.
- 4.4 Health education in the workplace.
- 5 Occupational Ergonomics In Special Areas 12 hrs**
- 5.1 Industrial worker.
- 5.2 Agriculture and rural areas.
- 5.3 Working women.
- 5.4 IT – BPO industries.

D. Instructional Method And Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

E. Student Learning Outcomes

At the end of the semester the student will be able:

- To review the basics of Occupational health & Ergonomics
- To understand the problem associated with workplace.
- To clinically correlate the Occupational hazards and Ergonomics.

F. Recommended Study Material

Text Books

1. Benjamin O. ALLI, Fundamental Principles of Occupational Health and Safety, Second edition, ILOInternational Labour Organization 2008
2. M. Rabiul&Ahasan: Occupational Health, Safety and Ergonomic issues in small and medium- sized Enterprises in a Developing Country. OULU 2002
3. WaldemarKarwowski& William S. Marras: Occupational Ergonomics – Design and management of Work System; CRC Press, Boca Raton, FL, 1999
4. Occupational Health : A manual for primary health care workers; WHO-EM/OCH/85/E/L

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF COMPUTER APPLICATION

CA 224 Introduction to Web Designing

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective

The objective of the course is to provide basic understanding of designing professional web page templates with Markup Language Tags.

B. Methodology & Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

C. Learning Outcome:

Upon successful completion of the course, students will understand basic concepts of internet and web page designing and design and develop entire web sites using several web designing editors and HTML scripting language.

D. Outline of the Course

Sr. No.	Content
1	Overview of Internet and WWW, Basic elements of the Internet, Internet services, Internet Browsers and Servers, Hardware and Software requirements to connect to the internet, Internet Service Provider (ISP), Introduction to Internet Protocols
2	Introduction to Web Page, Web Site, Web Browser, Overview of HTML, Structure of HTML Documents, HTML comments
3	HTML Basics Tags :Paragraph Tags, Horizontal Rule Tag, Heading Tags, Block quote Tags, Address Tags, PRE Tag,
4	Other HTML tags:- Formatting tags , Marquee tag, DIV tag, and SPAN tag
5	HTML List & Hyperlink in HTML (text link, image link, email link and many more)
6	HTML Images:- tag and <map> tag
7	HTML Table
8	HTML Form
9	HTML Frames

Total Hours (Practical): 24

E. Recommended Study Material

Core Books

1. Harley Hahn: The Internet Complete Reference, 2 nd Edition, Tata McGraw-HILL Edition.
2. Thomas a Powell: The Complete reference HTML, 3rd Edition, McGraw Hill, 2001.
3. A. Whyte: Basic HTML, 2nd Edition, Payne-Gallway, Oxford, 2003.
4. Farrar: HTML Example book, BPB, 2007.

Reference Books

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4th revised edition, BPB Publication.
2. Jeremy Keith : HTML5 for Web Designers, A Book Apart Jeffrey Zeldmann, 2010
3. Peter Morville & Louis Rosenfeld, Information Architecture for WWW, 3rd Edition, O'Reilly Publication, 2006.

Web References

1. <http://www.w3schools.com/html/>[HTML notes]
2. <http://www.tutorialspoint.com/html/> [HTML Materials]
3. <http://www.tutorialspoint.com/html5/>[HTML5 notes]

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF COMPUTER APPLICATION

CA 225 Programming the Internet

Credits and Hours

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2		2	2
Marks	00	100	100	

A. Objective

The objective of this course is to provide working knowledge of design effective and attractive web pages using HTML5 and CSS.

Pre-requisite: Basic knowledge of HTML

B. Methodology & Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

C. Learning Outcome

Upon Successful completion of the course, Students will learn the concepts of CSS from basic to advance. Students will be able to design attractive web pages using HTML5 and CSS

D. Outline of the Course

Sr. No.	Content
1	HTML5: Introduction, New Elements, Semantics, Media: Audio and Video
2	Forms: Input Types, Form Elements, Form Attributes, Graphics: Canvas
3	Introduction to Cascading Style Sheet (CSS), CSS layout, CSS essentials,
4	CSS selectors, CSS Box Model and Wrapper class
5	CSS Backgrounds and CSS Borders
6	CSS Text and Fonts and CSS List
7	CSS Tables, CSS Display, CSS Navigation and CSS Button
8	CSS Image Gallery and CSS 3:- Transforms, Transitions, Animation

Total Hours (Practical): 24

F. Recommended Study Material

Core Books

1. Matthew MacDonald: HTML5: The Missing Manual, O'Reilly Media, August 2011.
2. Peter Gasston: The Book of CSS3: A Developer's Guide to the Future of Web Design, No Starch Press, April 2011.
3. Richard York: Beginning CSS: Cascading Style sheets for Web Design, Wrox Press (Wiley Publishing), 2014.

Reference Books

1. Ivan Bayross: Web Enabled Commercial Application Development using HTML, JavaScript, DHTML and PHP, 4th revised edition, BPB Publication.
2. David Mc Farland: CSS: The Missing Manual, O'Reilly, 2006.
3. Thomas Powell: HTML & CSS: The Complete Reference, Fifth Edition Paperback, 2010

Web References

1. http://www.w3schools.com/html/html5_intro.asp [HTML5 notes]
2. <http://www.w3schools.com/css/> [CSS notes]
3. http://www.tutorialspoint.com/internet_technologies/html.htm [HTML notes]
4. <http://james-star.com/answers/en/css3-hover-effect-transitions-transformations-and-animations/> [Animation in CSS]

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES

BM231: BANKING AND INSURANCE (B & I)

(Choice Based Credit System – University Elective)

YEAR 2, SEMESTER 3

A. Number of Credits : 2

B. Course Objectives

The objectives of this course are:

- To equip the students with the knowledge of basic banking operation and Insurance industry.
- To recognize opportunities brought about by the dramatic changes that have occurred in the past decade in the banking and insurance industry.

C. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	Evolution of Banking • <i>Brief Structure of Banks</i> • <i>Systems of Banking-Mixed, Branch, Unit, Group, Chain</i> • <i>RBI-Organization & Functions</i> • <i>Methods of Credit Control & Credit Creation.</i> • <i>Commercial Banking</i> • <i>Universal Banking</i>	5
2	Bank Management • <i>Sources and Uses of Funds in Banks</i> • <i>Balance sheet of a Bank</i> • <i>Value Chain Analysis in Banking Industry</i> • <i>Emerging trends in Banking</i> • <i>Credit Cards</i> • <i>E-Banking</i>	5
3	Retail Banking – An Introduction • <i>Open Market Conditions and Role of Banks and Financial Institutions</i> • <i>Retail Banking – Concept and Importance.</i> • <i>Retail Banking Products- Housing Loan, Conveyance Loan, Personal Loan, Educational Loan, Loan for Retail Traders</i> • <i>Plastic Money</i>	5
4	Marketing of Banking Services – Banking Products and Services • <i>Distribution, Pricing and Promotion Strategy for Banking Services</i> • <i>Attracting and Retaining Bank Customers</i>	5

Module No.	Title/Topic	Classroom Contact Sessions
	Marketing Strategy of Credit Cards, Debit Cards, Saving Accounts and Different Types of Loans	
5	Introduction to Insurance Sector - Concept of Risk: Types of Risk, Risk Appraisal, Transfer and Pooling of Risks, Concept of Insurable Risk. - Concept of Insurance: Relevance of Insurance - Types of Insurance Organisations - Intermediaries in Insurance Business - Formation of Insurance Contract: Life, Fire, Marine and Motor Insurance Contracts - Principles of Insurance: Utmost Good Faith, Indemnity, Insurable Interest.	5
6	Practice of Life Insurance - Insurance Products, a Hedge Against Personal Risk (s) - Insurance Products, Alternative to Investment Products - Insurance Products, Collateral Security in the Rising Hire-Purchase Market Scenario - Marketing of Insurance Products- Life and Non Life Products - I.T in Insurance Business: Internet Based Delivery of Insurance Products, Servicing of Policies	5
	Total	30

D. Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may befit the course structure.

E. Learning Outcomes

At the end of the course, the student should have learnt

- Knowledge about various functions associated with banking.
- Practice and procedures relating to deposit and credit, documentation, monitoring and control.
- An insight into banking services and banking technology.
- Understanding about the insurance sector and its products.

G. Recommended Study Material

Text-Book

1. BhartiPathak, Indian Financial System, Pearson Education, Latest Edition
2. Gupta, P K. , Fundamentals of Insurance, Himalaya Publishing House, Latest Edition

Reference-Books

1. Latest Publications of Insurance Regulatory Authority of India
2. Khan M A, Introduction to Insurance, Educational Publication House, Aligarh, Latest Edition

Journals / Magazines / Newspapers

1. Journal on Banking Financial Services and Insurance Research
2. The Indian Banker

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES

BM241: HEALTHCARE MANAGEMENT (HCM)

(Choice Based Credit System – University Elective)

YEAR 2, SEMESTER 4

A. Number of Credits : 2

B. Course Objectives

The objectives of this course are:

- To help students understand, gain knowledge, and develop an appreciation of health care management by exploring all aspects of the field.
- To provide a foundation of applying managerial knowledge within health care sector.

C. Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	Hospital Planning and Medical Care • Steps Involved in Hospital Planning • Managing Patient Care Service • Managing Support Services • Administration and Engineering Service	5
2	Health Sector Financing and Reforms • Health Care Financing in India and Developed Nations • National Health Spending • Health Sector Reforms • Resource Generation for Hospitals • Hospital Financing Alternatives	5
3	Role of Private Sector in Health Care • Private Health Sector • Expanding the Private Sector Health Care Financing • Public- Private Mix in Health Care Sector • Legal Issues in Health Care Sector	5
4	Health Care Economics and Finance and Marketing • Economic Appraisal of Health Services • Needs Vs Demand Vs Supply Model • Health Service Financing and Expenditure Survey, Budgeting, Control, Pricing, and Efficiency • Production and Cost of Health Care • Hospital Costs and Efficiency Marketing Health and Family	5

Module No.	Title/Topic	Classroom Contact Sessions
	Welfare • Social Marketing for Health and Family Welfare • Marketing for Hospitals • Health Services Marketing; Pharmaceutical Marketing • Patient as a Customer	
5	Legal Aspects of Healthcare Management • Mitigate Liability Through Risk Management Principles • Develop Relationship Management Skills • Apply an Ethical Decision-Making Framework • Incorporate Employment Law Procedures • Manage Communication	5
6	Health Care and Social Policy • Social Welfare • Approaches to Analysis • Distribution of Health Services in India	5
	Total	30

D. Pedagogy

- University Elective Course shall be of 100 Marks. The Suggested evaluation component are as below;

Evaluation	Marks		Evaluation Methodology
External	70 Marks		Practical Examination, Presentation with Report, Case Studies, MOOCs, etc.
Internal	30 Marks	5 Marks for Attendance	N/A
		25 Marks for Continuous Evaluation	Quiz; Assignments, Seminars, Projects, Presentation, etc.

- The Pedagogy for application mode may include laboratory Sessions, Presentation, Case Studies and Discussions, MOOCs under staff observation or any adequate methodology which may benefit the course structure.

E. Learning Outcomes

At the end of the course, the student should have learnt:

- The relationship of social need for health care services and contemporary issues prevailing to it.

H. Recommended Study Material

Text-Book

1. S.L.Goel, Health Care Policies and Programmes, Deep and Deep Publications Pvt Ltd, Latest Edition

Reference-Books

1. G.D. Kunders, Designing for Total Quality in Health Care, Prism Books Pvt. Ltd., Bangalore, Latest Edition
2. B.M. Sakharkar, Principles of Hospital Administration and Planning, Jaypee Brothers Medical Publishers Pvt. Ltd., Latest Edition

Journals / Magazines / Newspapers

1. International Journal of Health Planning and Management
2. Health Policy and Planning Journal